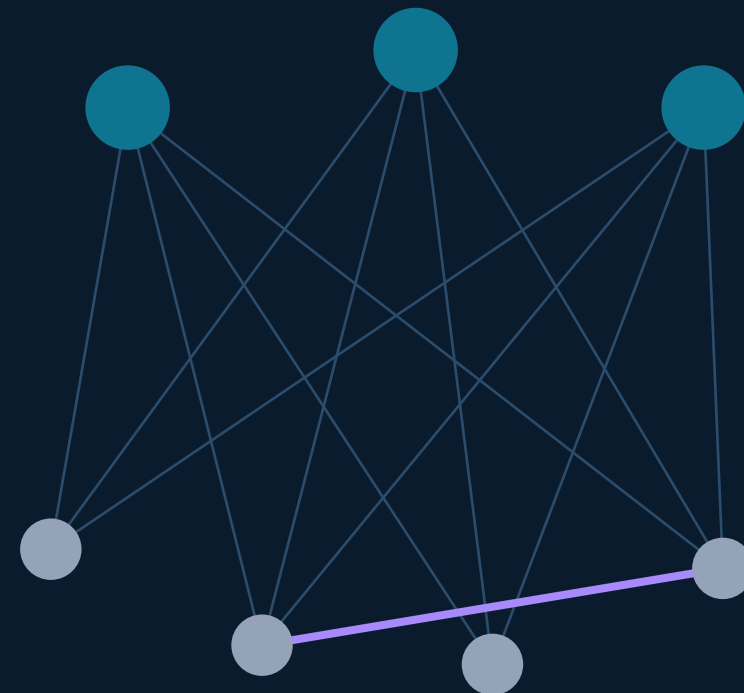


NETWORK AS CODE · DMVPN PHASE 3

C8000v DMVPN Portal

One place to see, change, prove and share a multi-hub DMVPN service:
for the operators who run it and the customers who use it.



Three Catalyst 8000v hubs · C8000v and VyOS customers · two providers · Nautobot · Terraform (NAC) · VictoriaMetrics

Version 0.21.3 · September 2026

Running a DMVPN service by hand does not scale

A single customer touches three hubs, one or two providers, two router operating systems and a source of truth. Every question and every change means logging in to several boxes.

Scattered state

Is every spoke registered with every hub? Are shortcuts forming? Health lives in a dozen CLI sessions on IOS XE and VyOS.

Slow, risky changes

Adding, changing or removing a customer means consistent edits on hubs, spoke, provider and Nautobot. It is easy to miss one.

Drift

Hand edits leave routers differing from the intended design, and nobody notices until something breaks.

No change control

Who changed what, who approved it, and what exactly changed on each router? Usually nobody can say.

Unproven resilience

Dual hubs and dual providers look good on paper, but how long does failover really take?

Blind customers

Customers cannot see their own service, test it, or ask for changes. SLA reports are assembled by hand.

Static secrets

The DMVPN pre-shared key is set once and rarely rotated, because rotating it by hand risks an outage.

The result

Slow onboarding, outages from avoidable mistakes, and no evidence for customers or auditors.

The diagram illustrates the architecture of the Portal :8094 system. At the center is the **Portal :8094**, which acts as a hub connecting various components:

- Operators**: Includes viewer, operator, approver, and admin roles.
- Customers**: Manages their own site, SLA, requests, and tests.
- lab.conf**: Represents the intended design.
- Render**: Acts as a hub for spoke and provider configuration.
- Nautobot**: Mirrors the model.
- Catalyst 8000v**: Uses Terraform (NAC) to generate RESTCONF configurations.
- VyOS**: Supports SSH push and declarative configurations.
- Providers**: Manages MPLS and MPLS2 (VyOS).
- VictoriaMetrics**: Monitors SLA and application probes.

Data Flow:

- Operators and Customers interact with the Portal :8094.
- The Portal :8094 interacts with lab.conf, Render, Nautobot, and VictoriaMetrics.
- lab.conf feeds into Render.
- Render feeds into Catalyst 8000v, VyOS, and Providers.
- Nautobot feeds into VictoriaMetrics.

Read path: The portal polls every router (RESTCONF, CLI, VyOS op mode) and draws what is really there, not what should be.

Monitoring: Every minute: pings, app checks.

Capabilities at a glance

1 See

Live cloud table and KPIs, a topology map drawn from live state, per-customer details, path trace and 'watch the shortcut form'.

2 Change

Add, modify and remove customers, deploy the model, fix drift, back up and restore, choose the preferred hub or add a second provider.

3 Control

Change requests with four-eyes approval, change windows, emergency changes, and maintenance mode that mutes alerts.

4 Audit

Every job snapshots every router before and after it runs, records who asked and who approved, and shows a diff per router.

5 Prove

SLA per customer (availability, latency, loss), application checks, simulated failures and measured failover times.

6 Secure

Sign-in with roles, admin-only pre-shared key rotation with a random key, and secrets masked in every view and log.

7 Customers

A read-only customer portal: its own site and applications, SLA and PDF report, change requests and self-service tests.

8 Automate

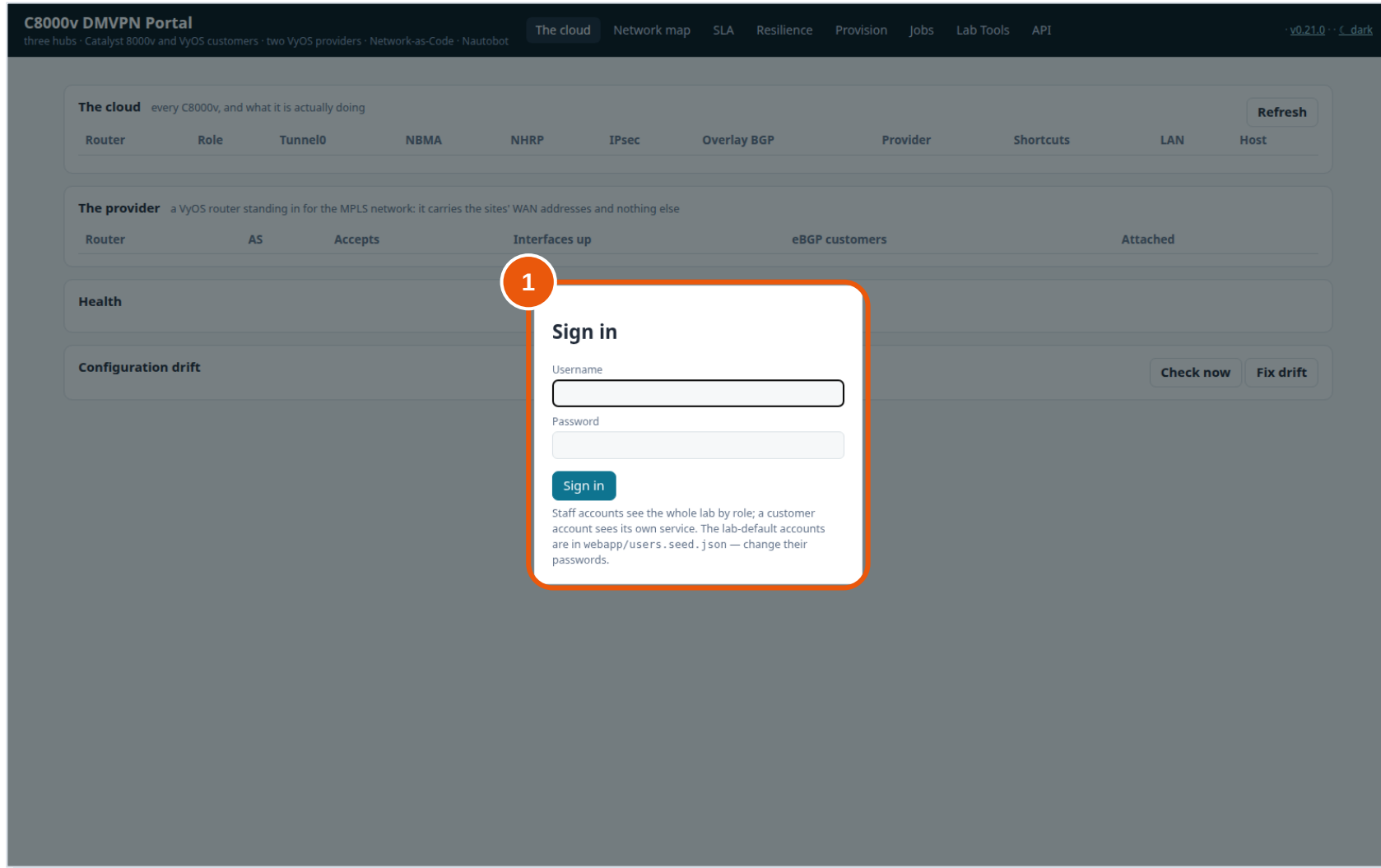
A REST API for everything (Swagger), Prometheus metrics, progress and ETA for each job, and a 73-test Robot suite after each change.



A tour of the portal

Numbered callouts on each screenshot match the explanations beside it.

Everyone signs in, and the role decides what they see



1

Sign in

Staff roles are viewer, operator, approver and admin. A customer account sees only its own service. Sessions are signed cookies, passwords are PBKDF2 hashes.

Why it matters

Read-only users cannot start jobs, nobody approves their own change, and customers never see another customer.

The cloud: every router, read live

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud

Network map

SLA

Resilience

Provision

Jobs

Lab Tools

API

administrator admin · admin · password · sign out v0.21.0 · 5 customers · 0 shortcuts · healthy · [dark](#)

1

3 Hubs

5 Customers

15 / 15 Registrations

0 Shortcuts live

65100 Overlay AS

65000 Provider AS

2

The cloud

every C8000v, and what it is actually doing

Refresh

Router	Role	Tunnel0	NBMA	NHRP	IPsec	Overlay BGP	Provider	Shortcuts	LAN	Host
hub-east East	hub	172.28.0.1	100.70.1.2	5 registered cloud 2: 2	11	11 / 11	both established	—	192.168.71.0/24	
hub-central Central	hub	172.28.0.2	100.70.2.2	5 registered cloud 2: 2	11	11 / 11	both established	—	192.168.72.0/24	
hub-west West	hub	172.28.0.3	100.70.3.2	5 registered cloud 2: 2	11	11 / 11	both established	—	192.168.73.0/24	
cust1 East Harborview Dental Group	customer	172.28.0.11	100.70.11.2	3 / 3 hubs cloud 2: 3 / 3	6	6 / 6	both established	none	192.168.61.0/24	host-cust1 up
cust2 Central Prairie Grain Logistics	customer	172.28.0.12	100.70.12.2	3 / 3 hubs	3	3 / 3	established	none	192.168.62.0/24	host-cust2 up
cust4 East Redwood Coast Brewing Co.	customer	172.28.0.14	100.70.14.2	3 / 3 hubs cloud 2: 3 / 3	8	6 / 6	both established	none	192.168.64.0/24	host-cust4 up
cust5 Central Summit Ridge Property Management	customer	172.28.0.15	100.70.15.2	3 / 3 hubs	5	3 / 3	established	none	192.168.65.0/24	host-cust5 up
cust6 West Gulfstream Marine Services	customer	172.28.0.16	100.70.16.2	3 / 3 hubs	5	3 / 3	established	none	192.168.66.0/24	host-cust6 up

A customer registers with every hub, so NHRP should read 3 / 3. Shortcuts are customer-to-customer tunnels NHRP has resolved (phase 3); they come and go with traffic.

3

The provider

a VyOS router standing in for the MPLS network: it carries the sites' WAN addresses and nothing else

Router	AS	Accepts	Interfaces up	eBGP customers	Attached
mpls primary	AS 65000	100.70.0.0/16	9 up	8 / 8	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust1 (eth4), cust2 (eth5), cust5 (eth6), cust4 (eth7), cust6 (eth8)
mpls2 second: the backup cloud	AS 65010	100.71.0.0/16	6 up	5 / 5	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust4 (eth4), cust1 (eth5)

Health

everything the model expects is true

- 1

KPIs
Hubs, customers, registrations, shortcuts and BGP at a glance.
- 2

Every router
NHRP registrations, IPsec, overlay BGP, the provider and the preferred hub, read from the routers themselves.
- 3

Both providers
Access links and eBGP on mpls and mpls2.
- 4

Who you are
The account, its role and the portal version, always visible.

Why it matters
One screen answers 'is every spoke registered with every hub?' with no CLI logins.

The provider					
a VyOS router standing in for the MPLS network: it carries the sites' WAN addresses and nothing else					
Router	AS	Accepts	Interfaces up	eBGP customers	Attached
mpls primary	AS 65000	100.70.0.0/16	9 up	8 / 8	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust1 (eth4), cust2 (eth5), cust5 (eth6), cust4 (eth7), cust6 (eth8)
mpls2 second: the backup cloud	AS 65010	100.71.0.0/16	6 up	5 / 5	hub-east (eth1), hub-central (eth2), hub-west (eth3), cust4 (eth4), cust1 (eth5)

Health

Everything the model expects is true

Configuration drift checked 9/27/2026, 5:38:23 PM · took 18s · every 6 h while the lab runs

Check now Fix drift

no drift: every router runs what the model says Nautobot == lab.conf

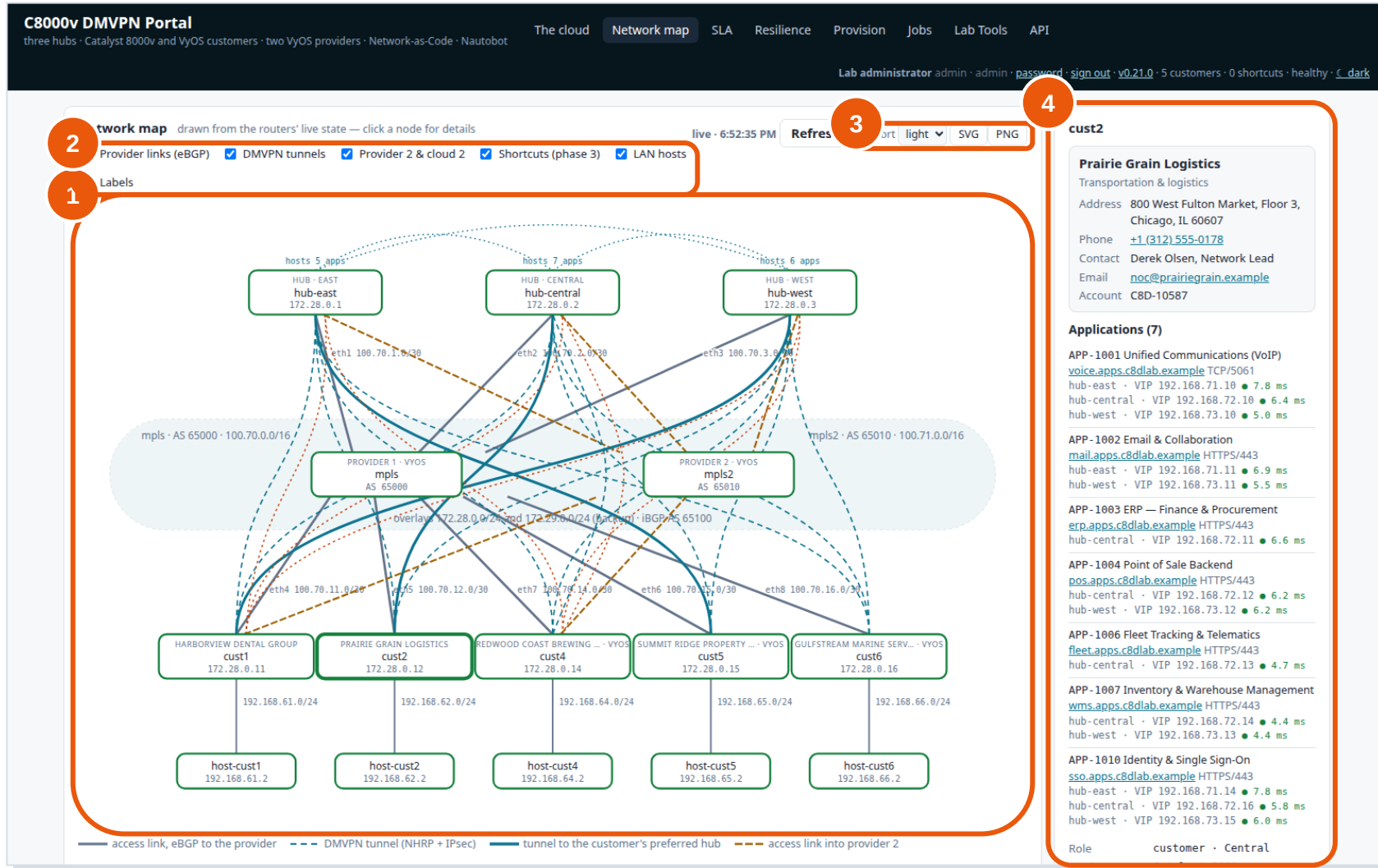
Router	Verdict	Differences (missing: the model has it, the router does not · extra: the router has it, the model does not · changed: Terraform would update it)
cust1 C8000v customer	matches the model	
cust2 C8000v customer	matches the model	
cust4 VyOS customer	matches the model	
cust5 VyOS customer	matches the model	
cust6 VyOS customer	matches the model	
hub-central C8000v hub	matches the model	
hub-east C8000v hub	matches the model	
hub-west C8000v hub	matches the model	
mpls VyOS provider	matches the model	
mpls2 VyOS provider	matches the model	

Everything the model expects to be up is checked, and anything missing is named.

Each router is compared with what lab.conf renders. 'Fix drift' puts it back as a controlled job.

Hand edits no longer go unnoticed: drift is visible per router, and fixing it is one approved job.

A topology map drawn from live state



1 The map
Hubs, providers, customers and LAN hosts. Tunnel colour shows health, and the preferred hub is drawn bold.

2 Layers
Turn provider 2 and cloud 2, shortcuts, hosts and labels on or off.

3 Export
SVG or PNG, light or dark, exactly as drawn.

4 Details
Click a node to see the customer company, its applications (checked every minute) and the router's live state.

Why it matters

Everyone gets the same live picture, from operations to account managers and customers.

Path trace: watch the phase 3 shortcut form

1

hosts 5 apps: HUB - EAST (172.28.0.1), HUB - CENTRAL (172.28.0.2), HUB - WEST (172.28.0.3)

hosts 7 apps: HUB - CENTRAL (172.28.0.2)

hosts 6 apps: HUB - WEST (172.28.0.3)

mpls AS 65000: 100.70.0.0/16

PROVIDER 1 - VYOS mpls AS 65000

PROVIDER 2 - VYOS mpls2 AS 65010

mpls2 AS 65010: 100.71.0.0/16

customer group 1: HARBORVIEW DENTAL GROUP (cust1: 172.28.0.11), PRAIRIE GRAIN LOGISTICS (cust2: 172.28.0.12), REDWOOD COAST BREWING ... (cust4: 172.28.0.14), SUMMIT RIDGE PROPERTY ... (cust5: 172.28.0.15), GULFSTREAM MARINE SERV... (cust6: 172.28.0.16)

customer group 2: host-cust1 (192.168.61.2), host-cust2 (192.168.62.2), host-cust4 (192.168.64.2), host-cust5 (192.168.65.2), host-cust6 (192.168.66.2)

Legend:

- access link, eBGP to the provider
- DMVPN tunnel (NHRP + IPsec)
- tunnel to the customer's preferred hub
- access link into provider 2
- cloud 2 tunnel (backup)
- live customer-to-customer shortcut
- down
- traced path through a hub
- configuration drift
- traced path over the shortcut
- outline: healthy · partial · down / unreachable

2

cust4

Address 300 Harbor Way, Eureka, CA 95501

Phone +1 (707) 555-0168

Contact Casey Lindqvist, Operations IT

Email it@redwoodcoast.example

Account C8D-10692

Role host

Platform Alpine Linux

Status healthy

VM c8d-host-cust4

Management 10.5.0.44

LAN address 192.168.64.2/24

Gateway 192.168.64.1 (cust4)

Login lab / lab

show configuration

ping all hosts (4)

connect to another host (traceroute, drawn on the map)

host-cust5 (cust5) trace path

watch the shortcut form clear path

3

host-cust4 → host-cust5: watching the shortcut form 6:53:00 PM direct (shortcut)

```
# clearing the shortcut between cust4 and cust5...
# cust4: clear ip nhrp cache (VyOS: every dynamic entry)
# cust5: clear ip nhrp cache (VyOS: every dynamic entry)
```

Copy Download Close

- 1 First via a hub, then direct**
The first packets go through a hub. NHRP then builds a spoke-to-spoke shortcut, drawn in purple.
- 2 Details and actions**
Trace, watch the shortcut form, ping every host, and view the config history per router.
- 3 Every trace, hop by hop**
The raw output is kept so you can copy or download it.

Why it matters

It shows that DMVPN phase 3 works end to end, and makes it easy to explain.

A router's configuration history

1 Router details

Its live state: addresses, NHRP, IPsec, BGP, and its actions.

BMA (Gi2)	100.70.1.2
Router-id	10.255.5.1
LAN	192.168.71.0/24
Provider link	mpls eth1 · 100.70.1.0/30
Provider 2 link	mpls2 eth1 · 100.71.1.0/30
Tunnel1 (cloud 2)	172.29.0.1 · NBMA 100.71.1.1
Registered	cust1, cust2, cust4, cust5, cust6
Registered (cloud 2)	cust1, cust4
IPsec up	11 / 11
Overlay BGP	11 / 11
Provider eBGP	both Established
CPU (1 min)	1 %

show dmvpn show crypto session brief

show ip bgp summary show ip route bgp

show ip nhrp show configuration

configuration history

2 hub-east - configuration history 2 job(s) changed it

Copy Download Close

```

2026-09-27_17-20-49-807a +1 / -1
@@ -263,5 +263,5 @@
peer ANY
address 0.0.0.0 0.0.0.0
- pre-shared-key <secret ...d76a>
+ pre-shared-key <secret ...85ff>
!
!

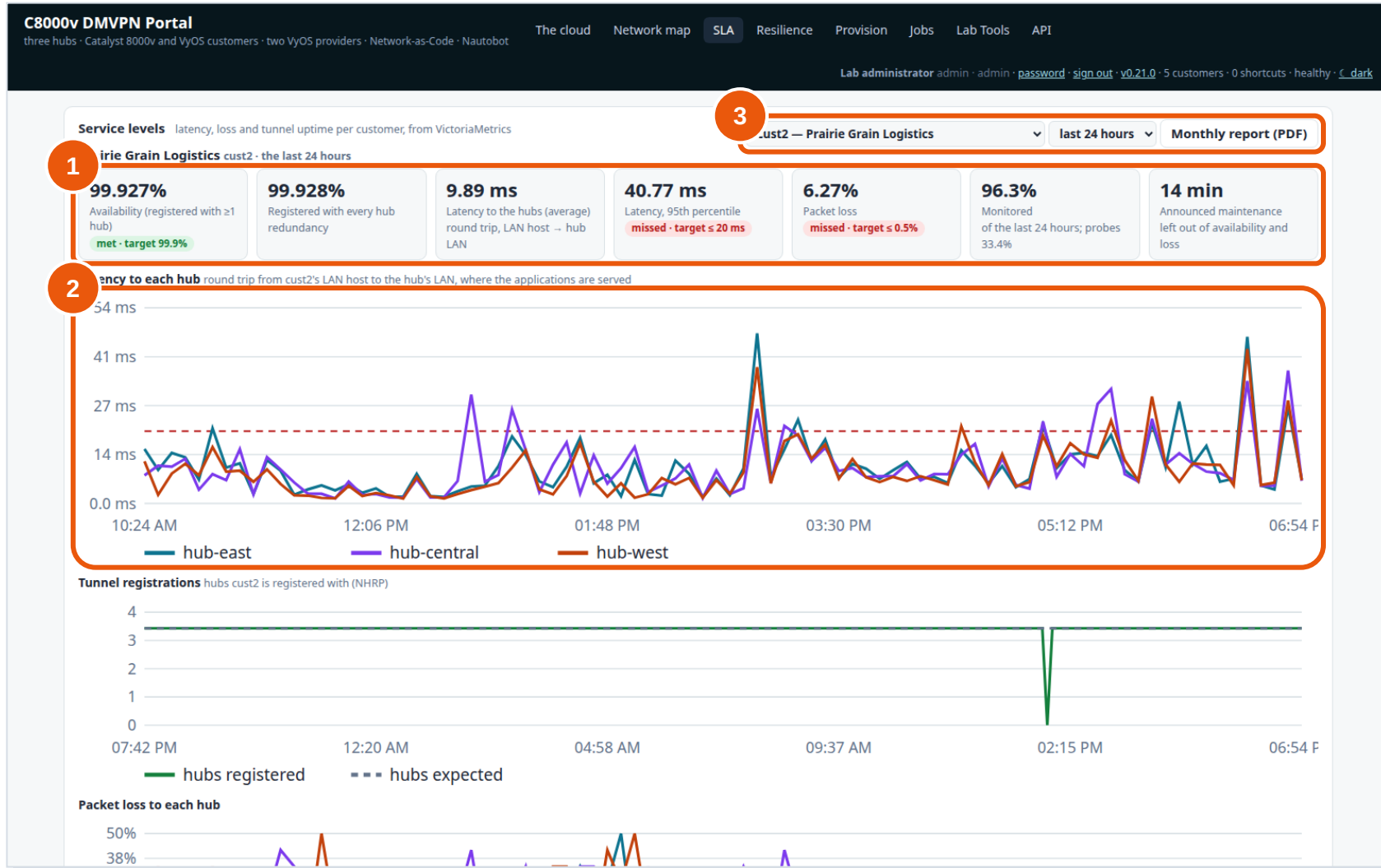
2026-09-27_15-01-20-1897 +5 / -0
@@ -223,4 +223,9 @@
interface Loopback10
description hub LAN
+ ip address 192.168.71.10 255.255.255.255 secondary
+ ip address 192.168.71.11 255.255.255.255 secondary
+ ip address 192.168.71.12 255.255.255.255 secondary
+ ip address 192.168.71.13 255.255.255.255 secondary
+ ip address 192.168.71.14 255.255.255.255 secondary
ip address 192.168.71.1 255.255.255.0
!

```

Why it matters

When something changed on a router, you can see which job did it and exactly what it changed.

SLA per customer, from VictoriaMetrics



1 Against targets
Availability, latency, loss and tunnel uptime, each compared with its SLA target.

2 Latency to each hub
Probed every minute from the customer's own LAN host.

3 Window and report
24 hours, 30 days or this month, with a monthly PDF report for the customer.

Why it matters

SLA reporting is automatic and based on measurements the customer can check.

Simulate failures on purpose

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloudNetwork mapSLAResilienceProvisionJobsLab ToolsAPI

Lab administratoradmin · admin · password · sign out · v0.21.0 · 5 customers · 0 shortcuts · healthy · [C_dark](#)

1

Simulate a failure

take part of the lab down on purpose, see what the design does, put it back

What happens: the hub's VM is frozen: no packets, no BGP, no NHRP

What the design should do: customers keep two hubs; routes through it move to the others when BGP gives up on it (hold 9 s); live shortcuts do not need the hub and stay up

Fail it now

Measure failover

In now

a failure left in is put back after 30 min

nothing is failed right now

Restore all

3

Failover measurements

every LAN host pings every other host and every hub's LAN five times a second while the fault is in

When	Failure	Held	Flows	Worst failover	Back to healthy
Sep 27, 01:49:13 PM	Hub fails on hub-east	45 s	29 unaffected 1 failed over 5 cut off	0.6 s	56.1 s
Sep 27, 01:44:24 PM	Provider fails on mpls	60 s	8 failed over 27 cut off	48.6 s	168.4 s
Sep 27, 01:34:12 PM	Provider fails on mpls	60 s	4 failed over 31 cut off	6.8 s	165.8 s

Earlier failures

Failure	Where	In	Out	How it ended
Customer circuit cut	cust1:wan2	Sep 27, 05:38:36 PM	Sep 27, 05:38:49 PM (13 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:53:23 PM	Sep 27, 03:53:34 PM (11 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:36:27 PM	Sep 27, 03:36:38 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:26:14 PM	Sep 27, 03:26:26 PM (11 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 03:21:17 PM	Sep 27, 03:21:27 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:45:05 PM	Sep 27, 02:45:14 PM (9 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:22:14 PM	Sep 27, 02:22:24 PM (10 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:12:08 PM	Sep 27, 02:12:17 PM (9 s)	restored
Customer circuit cut	cust1:wan2	Sep 27, 02:04:58 PM	Sep 27, 02:05:09 PM (11 s)	restored

- 1

Choose a failure

A hub, a provider, or a customer circuit. The page says what should happen before you press anything.
- 2

Measure failover

Every LAN host pings every other host five times a second while the fault is in.
- 3

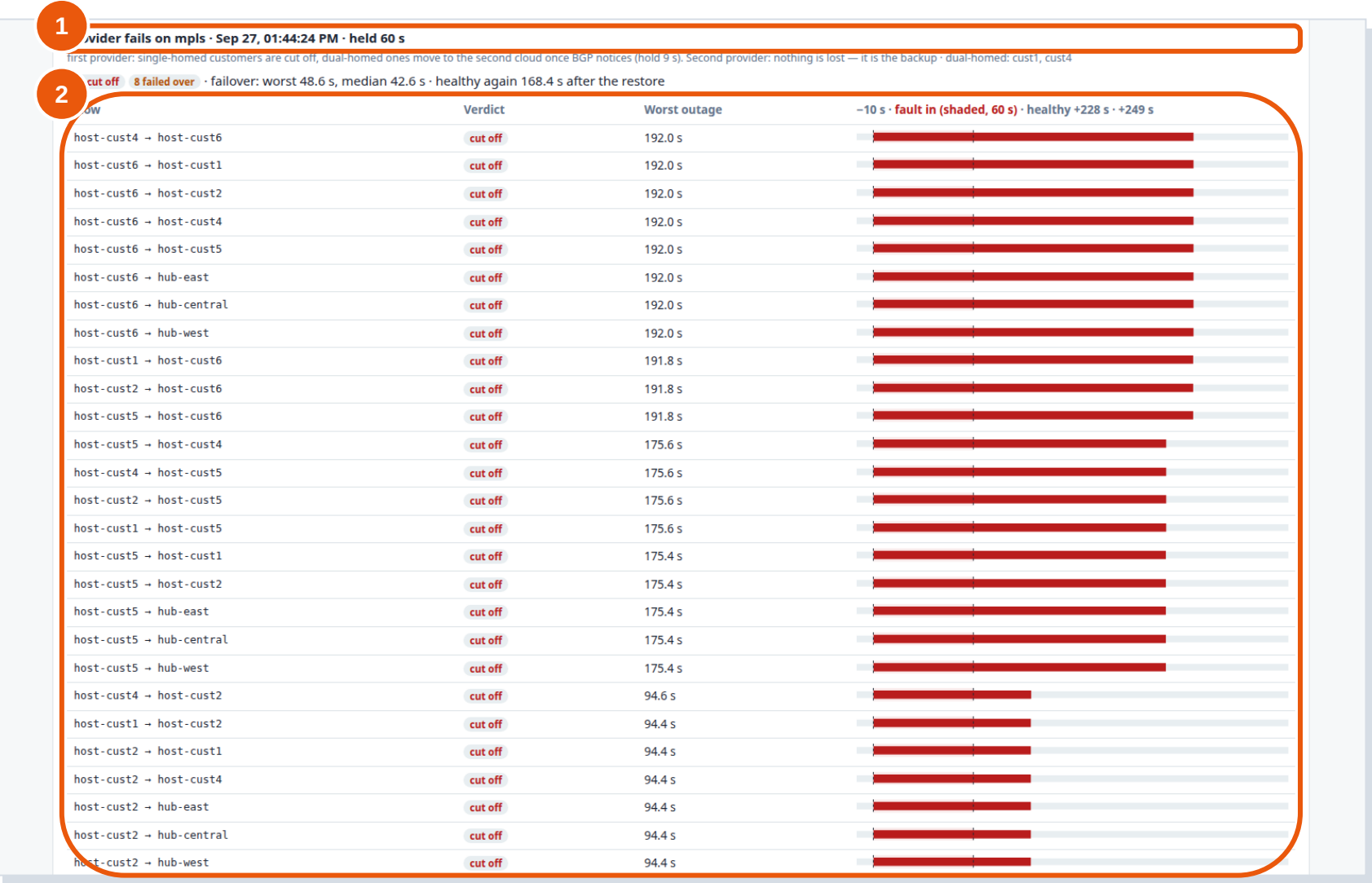
Every measurement

Flows unaffected, failed over or cut off, the worst failover and the time back to healthy.

Why it matters

Resilience becomes a number rather than a claim, for example 0.6 s worst case for a hub failure.

One failover experiment, flow by flow



- 1

The experiment
What failed, for how long, and the headline figures.
- 2

Each flow
The verdict and an outage timeline for every host pair, so you can see which paths moved and which were cut.

Why it matters
You can see exactly which customers a failure affects, and for how long.



Provision in depth

Every change is a job: planned, approved, run, verified and recorded.

- ## Tasks

Eleven tasks: each shows what it does, how long it takes and who may run it. The next slides walk through them.
- ## Pre-shared key

Admins rotate the DMVPN key. The system picks a random 40-character key, and only its fingerprint is ever shown.

Operators pick what they want done, and the portal carries it out consistently on every router.

Every Provision task, at a glance

Task	What it does	Time	Who	Approval
Add a customer	A new site: router (C8000v or VyOS) and LAN host, provider link, tunnels to all three hubs, Nautobot. No hub changes.	~15 min	operator	no
Modify a customer	Company, applications, preferred hub, site LAN, second provider, or router type (rebuilds the router).	1–15 min	operator	yes
Remove a customer	Terraform forgets it, VMs deleted, out of lab.conf and Nautobot, provider port released.	~3 min	operator	yes
Deploy the model	Re-render from lab.conf, terraform apply, push providers and VyOS, seed Nautobot, compare.	~4 min	operator	yes
Dry run	terraform plan and Nautobot's rendering compared with lab.conf. Changes nothing.	read only	operator	no
Check for drift	Every router's running configuration compared with the model.	1–2 min	operator	no
Fix drift	Put every router back to the model, then check again.	a few min	operator	yes
Back up the lab	One .tar.gz: lab.conf, customers, renders, Nautobot, running configs, Terraform state.	~30 s	operator	no
Restore from a backup	Upload, see exactly what would change, then rebuild the lab to it.	varies	operator	yes
Rotate the pre-shared key	A random 40-character key on every router; every IKE session re-made and verified.	~7 min	admin	yes
Run the tests	The Robot Framework suites against the live lab, with the evidence kept.	~10 min	operator	no

Approval follows the change policy (Jobs → Change policy...): four eyes, optional change windows, emergencies with a reason. Viewers and customers never start a job.

Add a customer: everything allocated for you

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS customers

What do you want to do? every change is made with the new key. The key never changes.

Add a customer
A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes. about 15 minutes (C8000v first boot)

Check for drift
Compare every router's running configuration with the model: Nautobot vs lab.conf, terraform plan for the C8000vs, the rendered set lines for VyOS. Changes nothing. read only · a minute or two

Run the tests
The Robot Framework suites against the live lab, with the evidence kept. about 10 minutes

DMVPN pre-shared key the IKEv2 key, as made with the new key. The key never changes.

Add a customer

1 Router type
Cisco Catalyst 8000v (IOS-XE, Network Services Engine) (any hub)

2 Preferred hub
none (any hub)

3 Dual-homed
☐ dual-homed (mpls2 eth6, 100.71.17.0/30)

4 Name and addresses
Index: cust7, Region: East, Management address: 10.5.0.27, Site LAN: 192.168.67.0/24

5 Index, port, host
Index: eth9, Provider port: eth9, LAN host: host-cust7, Host management: 10.5.0.47

6 The customer company (fictional, pre-filled — edit freely)
Company: Piedmont Family Pharmacy, Industry: Healthcare — retail pharmacy, Contact: Riley Johnson, Pharmacy Systems Lead

Address: 215 North Tryon Street, Charlotte, NC 28202
Phone: +1 (704) 555-0119
Email: it@piedmontfamily.example

Applications (what the customer subscribes to)

☐ APP-1001 Unified Communications (VoIP) ☒ APP-1002 Email & Collaboration ☐ APP-1003 ERP — Finance & Procurement

☐ APP-1004 Point of Sale Backend ☐ APP-1005 Electronic Health Records ☐ APP-1006 Fleet Tracking & Telematics

☐ APP-1007 Inventory & Warehouse Management ☐ APP-1008 Secure File Transfer ☐ APP-1009 Video Surveillance (NVR)

☒ APP-1010 Identity & Single Sign-On

Company Piedmont Family Pharmacy (Healthcare — retail pharmacy)

Router a new Catalyst 8000v router (10.5.0.27, console 5517, 4 GB, 2 vCPU; first boot ~10 min) and an Alpine host (10.5.0.47)

Cloud Tunnel0 172.28.0.17 sourced from 100.70.17.2, registered with hub-east, hub-central, hub-west

Hub no preferred hub: any of the hubs

WAN GigabitEthernet2 into mpls eth9 on 100.70.17.0/30

LAN 192.168.67.0/24 on GigabitEthernet3, with host-cust7 behind it

Changes the new router, mpls, Nautobot

the hubs: a customer registers with NHRP and arrives on their BGP listen range, so no hub configuration changes — Terraform adds resources on the new router only

☒ run the suites at the end

Provision the customer **Re-check** **Cancel**

- Router type**
Catalyst 8000v (NAC) or VyOS.
- Preferred hub**
Or none: any hub.
- Dual-homed**
A backup link into mpls2 and cloud 2.
- Name and addresses**
The next free name, management IP and LAN.
- Index, port, host**
Tunnel index, provider port, LAN host.
- The company**
Pre-filled; goes to Nautobot as a tenant.

Why it matters

Nothing to look up or calculate: every address and port is allocated and checked free.

Add a customer: the plan, before anything runs

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS

What do you want to do? every change

Add a customer
A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes. about 15 minutes (C8000v first boot)

Check for drift
Compare every router's running configuration with the model: Nautobot vs lab.conf, terraform plan for the C8000vs, the rendered set lines for VyOS. Changes nothing. read only · a minute or two

Run the tests
The Robot Framework suites against the live lab, with the evidence kept. about 10 minutes

DMVPN pre-shared key the IKEv2 key
Key secrets/dmvpn_psk
Set Sep 27, 05:20:53 PM
Used by hub-east, hub-central, hub-west
A rotation picks a random 40-character key, appears here, in the logs or in git: it is in secrets

Rotate the pre-shared key
A new random IKE pre-shared key on every router (it never leaves secrets/ on the lab host), then every router's IKE sessions set up again and checked. admin · about 7 minutes

Add a customer

Router type: **VyOS (FRR nhrpd + strongSwan)** Preferred hub: **none (any hub)** Second provider: ☒ **dual-homed** (mpls2 eth6, 100.71.17.0/30)

Name: **cust7** Region: **East** Management address: **10.5.0.27** Site LAN: **192.168.67.0/24**

Index (WAN /30, tunnel, router-id): **17** Provider port: **eth9** LAN host: **host-cust7** Host management: **10.5.0.47**

The customer company (fictional, pre-filled — edit freely)

Company: **Piedmont Family Pharmacy** Industry: **Healthcare — retail pharmacy** Contact: **Riley Johnson, Pharmacy Systems Lead**

Address: **215 North Tryon Street, Charlotte, NC 28202** Phone: **+1 (704) 555-0119** Email: **it@piedmontfamily.example**

1 Applications (what the customer subscribes to)

☒ **APP-1001** Unified Communications (VoIP) ☒ **APP-1002** Email & Collaboration ☐ **APP-1003** ERP — Finance & Procurement

☐ **APP-1004** Point of Sale Backend ☐ **APP-1005** Electronic Health Records ☐ **APP-1006** Fleet Tracking & Telematics

☐ **APP-1007** Inventory & Warehouse Management ☐ **APP-1008** Secure File Transfer ☐ **APP-1009** Video Surveillance (NVR)

☐ **APP-1010** Identity & Single Sign-On

2 Company **Piedmont Family Pharmacy (Healthcare — retail pharmacy)**

Router a new VyOS router (10.5.0.27, console 5517, 1 GB, 1 vCPU; first boot ~2 min) and an Alpine host (10.5.0.47)

Cloud Tunnel0 172.28.0.17 sourced from 100.70.17.2, registered with hub-east, hub-central, hub-west

Hub no preferred hub: any of the hubs

WAN GigabitEthernet2 into mpls eth9 on 100.70.17.0/30; and port 4 into mpls2 eth6 on 100.71.17.0/30 (dual-homed: Tunnel1 in the second cloud, the backup path)

LAN 192.168.67.0/24 on GigabitEthernet3, with host-cust7 behind it

Changes the new router, mpls, mpls2, Nautobot

the hubs: a customer registers with NHRP and arrives on their BGP listen range, so no hub configuration changes — Terraform adds resources on the new router only

3 run the suites at the end

provision the customer Re-check Cancel

- 1 Applications**
What the customer subscribes to; application checks start for them.
- 2 The plan**
Router, cloud, hub, WAN, LAN and what changes: here a new VyOS router, both providers and Nautobot. No hub changes.
- 3 Provision**
Starts the job, or files a change request if the policy asks for approval.

Why it matters

Onboarding a customer takes one form and about 15 minutes, with the whole impact visible up front.

Modify a customer, with the plan shown first

What do you want to do?

every change

Add a customer

A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes. about 15 minutes (C8000v first boot)

Check for drift

Compare every router's running configuration with the model: Nautobot vs lab.conf, terraform plan for the C8000vs, the rendered set lines for VyOS. Changes nothing. read only - a minute or two

Run the tests

The Robot Framework suites against the live lab, with the evidence kept. about 10 minutes

DMVPN pre-shared key

the IKEv2 key

Key	secrets/dmvpn...
Set	Sep 27, 05:20:53 PM
Used by	hub-east, hub-cent

A rotation picks a random 40-character key, appears here, in the logs or in git: it is in secret

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud · Network map · SLA · Resilience · Provision · Jobs · Lab Tools · API

Modify a customer

Customer

cust2 — Prairie Grain Logistics (Catalyst 8000v, C

1

2

3

4

Router type

Preferred hub

Second provider

The plan

Cisco Catalyst 8000v

hub-central (Central

second provider

single-homed → dual-homed (mpls2 eth6, 100.71.12.0/30)

Management address

Index (WAN /30, tunnel, router-id)

Provider port

LAN host

Host management

10.5.0.22

12

mpls eth5

host-cust2

10.5.0.42

The customer company

Company

Industry

Contact

Prairie Grain Logistics

Transportation & logistics

Derek Olsen, Network Lead

Address

Phone

Email

800 West Fulton Market, Floor 3, Chicago, IL 60607

+1 (312) 555-0178

noc@prairiegrain.example

Applications (what the customer subscribes to)

☒ APP-1001 Unified Communications (VoIP)

☒ APP-1002 Email & Collaboration

☒ APP-1003 ERP — Finance & Procurement

☒ APP-1004 Point of Sale Backend

☐ APP-1005 Electronic Health Records

☒ APP-1006 Fleet Tracking & Telematics

☒ APP-1007 Inventory & Warehouse Management

☐ APP-1008 Secure File Transfer

☐ APP-1009 Video Surveillance (NVR)

☒ APP-1010 Identity & Single Sign-On

The job

• cust2 is restarted with its port 4 wired into mpls2 eth6 (its configuration is saved first; a C8000v is back in ~5 min, a VyOS in ~1); then Tunnel1 registers with every hub in the second cloud

• Network-as-Code applied (C8000v) and the VyOS routers re-pushed; the hubs do not change

• Nautobot re-modelled: the customer is taken out and seeded again

Re-wiring port 4 restarts the router: the site is offline for a few minutes.

☒ run the suites at the end

Apply the changes

Re-check

Cancel

- 1

Router type
Move a customer between C8000v and VyOS.
- 2

Preferred hub
Pick the hub its traffic should prefer (local-preference 200).
- 3

Second provider
Dual-home the customer onto mpls2 and cloud 2.
- 4

The plan
Exactly what will change on which routers, before anything runs.

Why it matters
No surprises: the impact is visible and reviewed before a change request is filed.

- 1 Which customer**
Its router type, region and addressing are shown.
- 2 What goes with it**
The company and its subscriptions are removed from customers.json and Nautobot.
- 3 The plan**
The router and host deleted, Terraform state, the provider port released, lab.conf and Nautobot updated.
- 4 Remove**
Destructive: it needs an approver under the default policy.

Decommissioning leaves nothing behind: no orphaned ports, config, Terraform state or Nautobot records.

Deploy the model: lab.conf onto every router

The screenshot shows a deployment interface with a table of jobs. The job 'cust5 VyOS' is highlighted, showing a success status and a progress bar. Below the table, a detailed log for the job is displayed, showing the steps taken to deploy the model to the routers.

1 Deploy by operator, approved by approver success

2 success Snapshot every router's configuration (before) every router saved

2. success Render the configuration from lab.conf configuration re-rendered from lab.conf

3. success Network-as-Code: terraform apply (resources on the new router only) applied and saved

4. success The provider: address the new access link; the customer arrives on its listen range provider re-pushed (idempotent; a changed port is reconciled)

5. success Nautobot source of truth (seed) seeded

6. success Compare Nautobot's rendering with lab.conf's Nautobot == lab.conf

7. success Snapshot every router's configuration (after) and compare cust4 +1/-1; cust5 +1/-1; cust6 +1/-1; hub-central +7/-0; hub-east +5/-0; hub-west +6/-0; mpls +1/-1; mpls2 +1/-1

Configuration changes: cust4 +1/-1 cust5 +1/-1 cust6 +1/-1 hub-central +7/-0 hub-east +5/-0 hub-west +6/-0 mpls +1/-1 mpls2 +1/-1 show the differences

```
[render] 0 of 11 file(s) changed
[mls] applied 76 lines, committed and saved
[mls2] applied 76 lines, committed and saved
[cust4] applied 171 lines, committed and saved
[cust4] 99-c8d-ike-noesn: installed and run
[cust5] applied 143 lines, committed and saved
[cust5] 99-c8d-ike-noesn: installed and run
[cust6] applied 122 lines, committed and saved
[cust6] 99-c8d-ike-noesn: installed and run
$ /home/dcantor/c8000v-dmvpn-lab/lab.sh nautobot seed
seed complete: 0 changes (already in sync)
$ /home/dcantor/c8000v-dmvpn-lab/lab.sh nautobot render --check
nac/data/devices.nac.yaml: Nautobot == lab.conf (1246 lines)
nodes/cust1/iosxe_config.txt: Nautobot == lab.conf (43 lines)
nodes/cust2/iosxe_config.txt: Nautobot == lab.conf (43 lines)
nodes/cust4/vyos_config.txt: Nautobot == lab.conf (193 lines)
nodes/cust5/vyos_config.txt: Nautobot == lab.conf (165 lines)
nodes/cust6/vyos_config.txt: Nautobot == lab.conf (142 lines)
nodes/hub-central/iosxe_config.txt: Nautobot == lab.conf (43 lines)
nodes/hub-east/iosxe_config.txt: Nautobot == lab.conf (43 lines)
nodes/hub-west/iosxe_config.txt: Nautobot == lab.conf (43 lines)
nodes/mls/vyos_config.txt: Nautobot == lab.conf (84 lines)
nodes/mls2/vyos_config.txt: Nautobot == lab.conf (84 lines)
```

1 The job
Who started it, who approved it, 7 of 7 steps, 4 minutes.

2 The steps
Snapshot, render, Terraform apply (NAC), the providers, Nautobot seeded, Nautobot's rendering compared with lab.conf, snapshot and compare. The log is kept below.

Why it matters

The same pipeline builds, fixes and restores the lab, so the result is always the model.

Restore from a backup: see the difference first

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud Network map SLA Resilience Provision Jobs Lab Tools API

Lab administrator admin · admin · password · sign out · v0.21.1 · 5 customers · 0 shortcuts · healthy · [dark](#)

What do you want to do? every change is a job; pick one, review what it will do, then watch it

Add a customer
A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes. about 15 minutes (C8000v first boot)

Modify a customer
Change a customer: its company details and configurations; its preferred hub; its site.

Remove a customer
Decommission a customer: Terraform destroys it; the VMs are deleted; out of

Deploy the model
Re-render from lab.conf, terraform apply to every C8000v; push the provider; send

Dry run
terraform plan against the routers, and Nautobot's rendering compared with lab.conf's. Changes nothing. read only

Check for drift
Compare every router's running configuration with the model: Nautobot lab.conf, terraform plan for the C8000v; the rendered set lines for VyOS. Change nothing. read only · a minute or two

Run the tests
The Robot Framework suites against the live lab, with the evidence kept. about 10 minutes

DMVPN pre-shared key the IKEv2 key

Key secrets/dmvpn_psk

Set Sep 27, 05:20:53

Used by hub-east, hub-central

Rotate the pre-shared key
A new random IKE pre-shared key on every router (it never leaves secrets/ on the lab host), then every router's IKE sessions set up again and checked. admin · about 7 minutes

Restore from a backup
A backup made by **Back up the lab**. Nothing changes until you press Restore: first you see what it would do.

1 **Upload file (.tar.gz)**

2 **Choose File** 20260927-103824-4970.tar.gz

3 **Backup** c8000v-dmvpn-lab v0.17.0 · 9/27/2026, 10:38:24 AM · from ubuntu · 28 files, every checksum verified

Its customers
cust1 (Harborview Dental Group, C8000v, 192.168.61.0/24, prefers hub-west) · cust2 (Prairie Grain Logistics, C8000v, 192.168.62.0/24) · cust4 (Redwood Coast Brewing Co., VyOS, 192.168.64.0/24) · cust5 (Summit Ridge Property Management, VyOS, 192.168.65.0/24, prefers hub-east) · cust6 (Gulfstream Marine Services, VyOS, 192.168.66.0/24)

Changed cust1 company / applications

Changed cust2 prefer_hub: hub-central → ; company / applications

Also DMVPN holdtime: 60 → 300

Then Network-as-Code applied, the provider and VyOS routers pushed, Nautobot seeded from the restored intent, the lab verified

4 **Restore the lab to this backup**

Cancel

A rotation picks a random 40-character key, applies it to every router (Network-as-Code, the VyOS push), sets every router's IKE sessions up again one router at a time, and verifies that every session was made with the new key. The key never appears here, in the logs or in git: it is in secrets/dmvpn_psk on the lab host, shown only by its fingerprint. An admin rotates it; change control applies.

- The backup**
A .tar.gz made by **Back up the lab**, uploaded here.
- What is in it**
Version, date, host, files, every checksum verified, and its customers.
- What would change**
Customers changed, added or removed, other settings, then NAC, pushes, Nautobot and verification.
- Restore**
Only after approval; nothing changes before.

Why it matters

A known-good state is always one approved job away, and you know exactly what it will undo.

The DMVPN pre-shared key

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud Network map SLA Resilience **Provision** Jobs Lab Tools API

Lab administrator admin · admin · [password](#) · [sign out](#) · v0.21.1 · 5 customers · 0 shortcuts · healthy · [C_dark](#)

What do you want to do? every change is a job: pick one, review what it will do, then watch it

- Add a customer**
A new site: a C8000v and its LAN host, a link into the provider, Tunnel0 registered with all three hubs. No hub changes.
about 15 minutes (C8000v first boot)
- Modify a customer**
Change a customer: its company details and applications, its preferred hub, its site LAN, or its router type (C8000v or VyOS).
data only: a minute · a new router: up to 15 minutes
- Remove a customer**
Decommission a customer: Terraform forgets it, the VMs are deleted, out of lab.conf and Nautobot, its provider port released.
about 3 minutes
- Deploy the model**
Re-render from lab.conf, terraform apply to every C8000v, push the provider, seed Nautobot and compare.
nothing is created
- Dry run**
terraform plan against the routers, and Nautobot's rendering compared with lab.conf's. Changes nothing.
read only
- Check for drift**
Compare every router's running configuration with the model: Nautobot vs lab.conf, terraform plan for the C8000vs, the rendered set lines for VyOS. Changes nothing.
read only · a minute or two
- Fix drift**
Put every router back to the model: re-render, terraform apply, re-push the VyOS routers, re-seed Nautobot — then check again.
a few minutes
- Back up the lab**
Download the whole lab state as one .tar.gz: lab.conf, customers and applications, every rendered file, this lab in Nautobot, the routers' running configurations, the Terraform state.
read only · half a minute
- Restore from a backup**
Upload a backup, see what it would change — customers removed, added, changed — and rebuild the lab to it.
[depends on the changes](#)
- Rotate the pre-shared key**
A new random IKE pre-shared key on every router (it never leaves secrets/ on the lab host), then every router's IKE sessions set up again and checked.
admin · about 7 minutes
- Run the tests**
The Robot Framework suites against the live lab, with the evidence kept.
about 10 minutes

1 VPN pre-shared key the IKEv2 key every router authenticates with

2 y secrets/dmvpn_psk (rotated) · fingerprint ..552085ff

3 t Sep 27, 05:20:53 PM (0 days ago) · by admin, approved by approver (job 2026-09-27_17-20-49-807a)

sed by hub-east, hub-central, hub-west, cust1, cust2, cust4, cust5, cust6

A rotation picks a random 40-character key, applies it to every router (Network-as-Code, the VyOS push), sets every router's IKE sessions up again one router at a time, and verifies that every session was made with the new key. The key never appears here, in the logs or in git: it is in secrets/dmvpn_psk on the lab host, shown only by its fingerprint. An admin rotates it; change control applies.

4 Rotate the pre-shared key

- 1 Key**
Where it is kept (secrets/ on the lab host) and its fingerprint. The key itself is never shown.
- 2 Set**
When it was last rotated, by whom, and the job.
- 3 Used by**
Every hub and customer router.
- 4 Rotate**
Admins only, with a confirmation, and through change control.

Why it matters

Rotating the key becomes routine: one button, a random key, and proof that every session uses it.

A job: who, what, and the proof

1 Rotate pre-shared key by admin, approved by approver success

2 100% - 9 / 9 steps took 5m 28s

4

- success** Snapshot every router's configuration (before)
every router saved
- 2. success** Choose a new random pre-shared key (kept in secrets/, never in git or the log)
new key ...552085ff (the key itself is in secrets/dmvpn_psk only)
- 3. success** Render the configuration from lab.conf
configuration re-rendered from lab.conf
- 4. success** Network-as-Code: terraform apply (resources on the new router only)
applied and saved
- 5. success** The provider: address the new access link; the customer arrives on its listen range
provider re-pushed (idempotent; a changed port is reconciled)
- 6. success** Re-authenticate every IKE session, one router at a time
every router's IKE sessions set up again
- 7. success** Verify every IKE session: how it authenticated, and that it is new
54 IKE SAs on 8 routers, all authenticated with the pre-shared key, all made after the rotation
- 8. success** Verify: registered with every hub, IPsec and BGP up, hosts reach each other, Nautobot == lab.conf
the cloud is healthy; every host reaches every other; Nautobot == lab.conf
- 9. success** Snapshot every router's configuration (after) and compare
cust1 +1/-1; cust2 +1/-1; cust4 +1/-1; cust5 +1/-1; cust6 +1/-1; hub-central +1/-1; hub-east +1/-1; hub-west +1/-1

Configuration changes: cust1 +1 / -1 cust2 +1 / -1 cust4 +1 / -1 cust5 +1 / -1 cust6 +1 / -1 hub-central +1 / -1 hub-east +1 / -1 hub-west +1 / -1 show the differences

3

```

--- cust1 before
+++ cust1 after
@@ -263,5 +263,5 @@
peer ANY
address 0.0.0.0 0.0.0.0
- pre-shared-key <secret ...d76a>
+ pre-shared-key <secret ...85ff>
!
!

▼ cust2 +1 / -1

--- cust2 before
+++ cust2 after
@@ -176,5 +176,5 @@
peer ANY
address 0.0.0.0 0.0.0.0
- pre-shared-key <secret ...d76a>
+ pre-shared-key <secret ...85ff>
!
!

▼ cust4 +1 / -1

```

- 1 Who**
Who started the job and who approved it.
- 2 Progress**
Live progress and the time it took.
- 3 What changed**
The before-and-after diff per router. Secrets are masked but their change is still visible.
- 4 Every step**
Each step's result, for example '54 IKE SAs on 8 routers, all made after the rotation'.

Why it matters

Every change leaves evidence: the steps, the approvals and the exact configuration difference.

Run the tests: the lab proves itself

9/27/2026, 12:37:13 PM	modify	cust1 C8000v	success	100% · 4 / 4 steps took 2m 06s
9/27/2026, 11:23:06 AM	test		success	100% · 1 / 1 steps took 8m 44s 59/59
9/27/2026, 11:16:42 AM	test		success	100% · 1 / 1 steps took 6m 13s 37/37
9/27/2026, 11:06:02 AM	test		failed	0% · 0 / 1 steps took 8m 23s 54/59
9/27/2026, 10:54:41 AM	restore	20260927-105435-801f	success	100% · 8 / 8 steps took 10m 06s
9/27/2026, 10:43:38 AM	modify	cust6 C8000v	success	100% · 7 / 7 steps took 10m 27s
9/27/2026, 10:38:33 AM	modify	cust6 VyOS	success	100% · 7 / 7 steps took 4m 53s
9/27/2026, 10:31:28 AM	modify	cust5 VyOS	success	100% · 6 / 6 steps took 3m 56s

1

2

3

Test by operator success

success Robot Framework suites
73/73 passed

Tests: 73/73 passed

```
73 tests, 73 passed, 0 failed
-----
Output: /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/output.xml
Log: /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/log.html
Report: /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/report.html
==> capturing router configurations (post-run backup)
[hub-east] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-east.running-config.txt (13095 bytes)
[hub-east] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-east.startup-config.txt (9290 bytes)
[hub-central] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-central.running-config.txt (13194 bytes)
[hub-central] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-central.startup-config.txt (9389 bytes)
[hub-west] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-west.running-config.txt (13145 bytes)
[hub-west] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/hub-west.startup-config.txt (9340 bytes)
[cust1] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/cust1.running-config.txt (13559 bytes)
[cust1] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/cust1.startup-config.txt (9758 bytes)
[cust2] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/cust2.running-config.txt (11381 bytes)
[cust2] /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/configs/post-run/cust2.startup-config.txt (7576 bytes)
[cust4] config captured
[cust5] config captured
[cust6] config captured
[mpls] config captured
no configuration changes during the run
==> report: /home/dcantor/c8000v-dmvpn-lab/results/2026-09-27_17-29-44/report.html (rc=0)
```

- 1

The job
Robot Framework suites against the live lab.
- 2

The result
73 of 73 passed, in about 11 minutes.
- 3

The evidence
The Robot report and log, and every router's configuration before and after.

Why it matters
Every change can end with a full test run, and the evidence is one click away.

Maintenance and the job list

1

Maintenance under way while a disruptive job runs or a failure is simulated — or declared here: the nodes' alerts are muted, the customers told, the SLA leaves it out

none under way

What

replacing hub-west's uplink

Affects

the whole lab

2

Declare maintenance

recent maintenance

3

one at a time; a failed job resumes from the step that failed

Started	Mode	Customer	Status	Progress	Tests
9/27/2026, 5:29:44 PM	test		success	100% · 1 / 1 steps took 10m 56s	73/73
9/27/2026, 5:20:49 PM	rotatepsk	pre-shared key	success	100% · 9 / 9 steps took 5m 28s	
9/27/2026, 3:44:50 PM	test		success	100% · 1 / 1 steps took 10m 24s	72/72
9/27/2026, 3:42:10 PM	test		success	100% · 1 / 1 steps took 2m 29s	11/11
9/27/2026, 3:39:05 PM	test		failed	0% · 0 / 1 steps took 2m 56s	17/18
9/27/2026, 3:28:09 PM	test		failed	0% · 0 / 1 steps took 10m 05s	69/70
9/27/2026, 3:23:31 PM	test		success	100% · 1 / 1 steps took 3m 40s	26/26
9/27/2026, 3:18:52 PM	test		failed	0% · 0 / 1 steps took 3m 17s	24/26
9/27/2026, 3:13:18 PM	modify	cust2 C8000v	success	100% · 8 / 8 steps took 4m 31s	
9/27/2026, 3:01:20 PM	deploy		success	100% · 7 / 7 steps took 4m 02s	
9/27/2026, 2:42:27 PM	test		success	100% · 1 / 1 steps took 3m 17s	21/21
9/27/2026, 2:13:56 PM	test		success	100% · 1 / 1 steps took 9m 47s	65/65
9/27/2026, 2:07:37 PM	test		success	100% · 1 / 1 steps took 6m 08s	37/37
9/27/2026, 1:56:37 PM	test		failed	0% · 0 / 1 steps took 9m 44s	62/65
9/27/2026, 1:48:56 PM	failover	hub-east	success	100% · 6 / 6 steps took 2m 09s	
9/27/2026, 1:44:08 PM	failover	mpls	success	100% · 6 / 6 steps took 4m 16s	

- 1

Maintenance

Declared here or started automatically by disruptive jobs and simulated failures. Alerts are muted, customers told, and SLA leaves it out.
- 2

Declare

What, and which nodes it affects.
- 3

Every job

Status, a progress bar, duration and test results. A failed job resumes from the step that failed.

Why it matters

Planned work does not page anyone or count against the SLA, and customers see it coming.



Change control

A second pair of eyes, change windows, and a record of everything.

Change requests: four eyes and change windows

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloudNetwork mapSLAResilienceProvisionJobsLab ToolsAPI

Lab administratoradmin · admin · password · sign out · v0.21.0 · 5 customers · 1 shortcuts · healthy · [C_dark](#)

Change requests

approval for: remove, modify, restore, deploy, fixdrift, failover, fault, rotatepsk · four eyes · windows off

Change policy...

1

Accounts still on their lab-default password: admin, approver, operator, viewer — change them in Lab Tools → Users.

Change	What	Asked by	Status
CR-0023 Sep 27, 05:39:15 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0022 Sep 27, 05:38:29 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0021 Sep 27, 05:20:49 PM	Rotate the pre-shared key rotate the IKE pre-shared key on every router why: first rotation off the public lab default	admin approved by approver: rotate	done job
CR-0020 Sep 27, 03:53:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0019 Sep 27, 03:53:20 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0018 Sep 27, 03:36:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0017 Sep 27, 03:36:24 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0016 Sep 27, 03:26:55 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0015 Sep 27, 03:26:11 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0014 Sep 27, 03:21:54 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected

1

Change requests

Each request shows who asked, what for, who approved or rejected it, and its status, including requests from customers.

Why it matters

Disruptive work cannot start without a second person, and it runs only inside the agreed windows.

The change policy, edited in the portal

C8000v DMVPN Portal
three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloud Network map SLA Resilience Provision **Jobs** Lab Tools API

Lab administrator admin · admin · password · sign out · v0.21.1 · 5 customers · 0 shortcuts · healthy · [C_dark](#)

Change request approval for: remove, modify, restore, deploy, fixdrift, failover, fault, rotatepsk · four eyes · windows off Change policy...

1 **2**

Accounts still on their lab-default password: admin, approver, operator, viewer — change them in Lab Tools → Users.

☒ **approval needed for:** ☒ Remove a customer ☒ Modify a customer ☒ Restore from a backup ☒ Deploy the model ☒ Fix drift ☒ Measure a failover ☒ Simulate a failure ☒ Rotate the pre-shared key

☒ four eyes: the approver is never the requester ☒ emergency approvals allowed (start now, outside the window, with a reason) requests expire after h

☐ **change windows:** approved changes start only inside one

☒ mon ☒ tue ☒ wed ☒ thu ☒ fri ☐ sat ☐ sun from to

☐ mon ☐ tue ☐ wed ☐ thu ☐ fri ☒ sat ☒ sun from to

Change	What	Asked by	Status
CR-0023 Sep 27, 05:39:15 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0022 Sep 27, 05:38:29 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0021 Sep 27, 05:20:49 PM	Rotate the pre-shared key rotate the IKE pre-shared key on every router why: first rotation off the public lab default	admin approved by approver: rotate	done job
CR-0020 Sep 27, 03:53:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0019 Sep 27, 03:53:20 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started
CR-0018 Sep 27, 03:36:59 PM	Modify a customer modify cust1: preferred hub hub-west → hub-east why: suite 07	robot-customer rejected by approver: suite 07: only a test	rejected
CR-0017 Sep 27, 03:36:24 PM	Simulate a failure simulate: Customer circuit cut on cust1:wan2 why: suite 07	operator approved by approver: suite 07	started

1

The policy

Which tasks need approval, four eyes, emergency approvals with a reason, expiry after 72 h, and change windows by day and time.

2

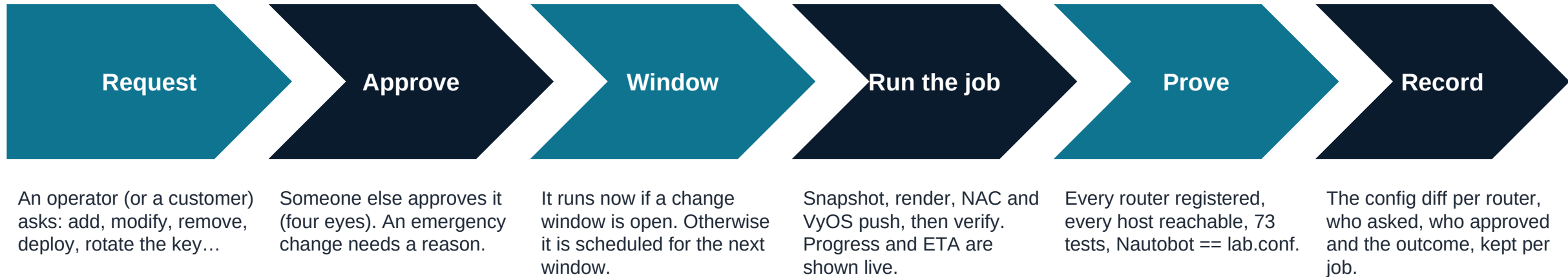
In one line

The current policy is always shown above the requests.

Why it matters

The rules for change are visible and adjustable, not tribal knowledge.

How a change moves through the portal



What is never allowed

- Approving your own request (four eyes)
- A viewer or customer starting a job
- Rotating the pre-shared key unless you are an admin
- A secret reaching git, a job log or the screen

What never needs approval

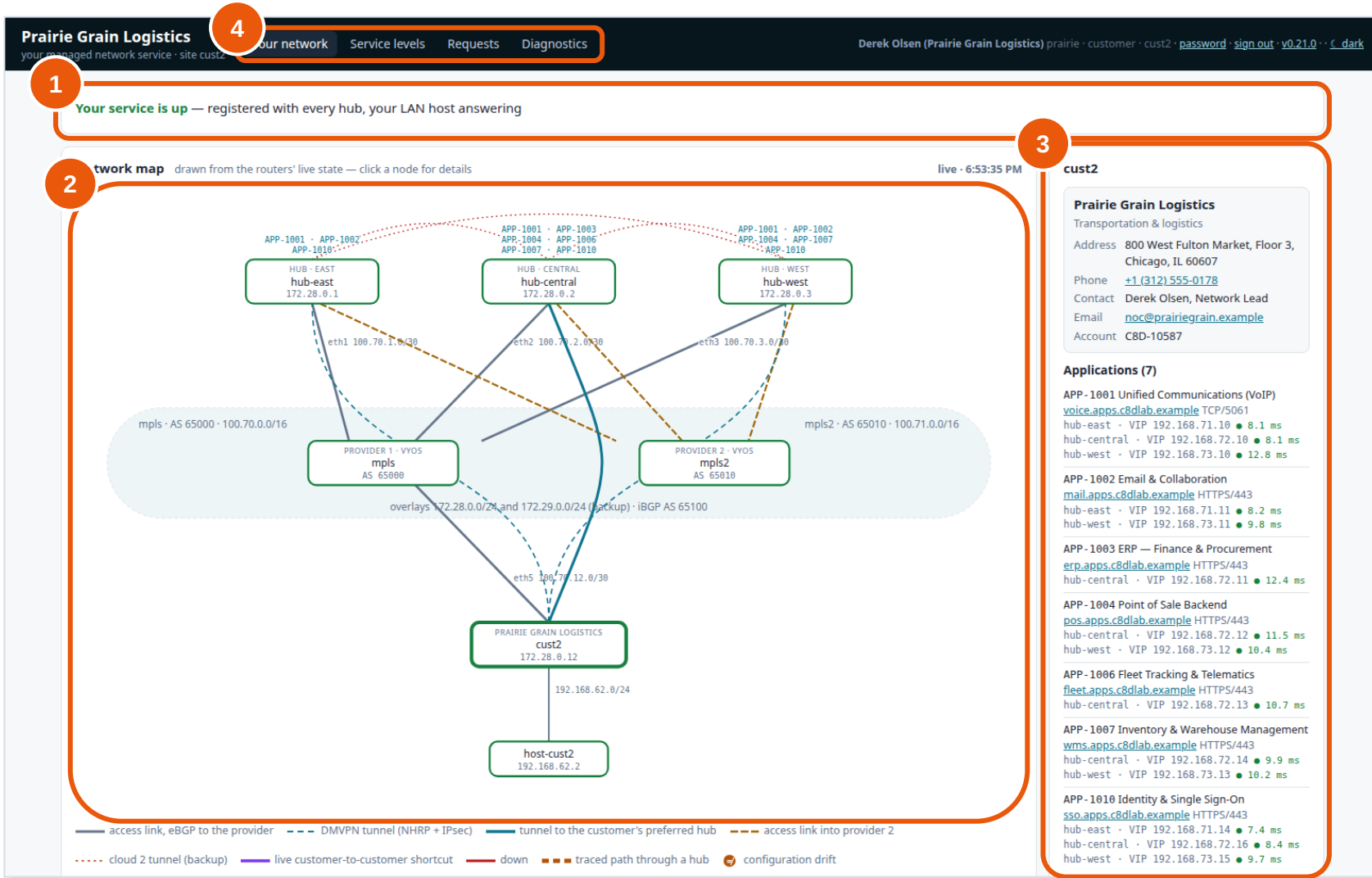
- Adding a customer: it takes nothing away
- Dry runs, tests and drift checks
- Customer diagnostics from its own LAN host
- Reading anything the role allows



Customers, access and the API

Each customer sees its own service; staff see what their role allows.

The customer's own view



- 1 Status**
Service status, incidents and planned maintenance for this customer.
- 2 Its own site**
The map, cut down to the customer's own site and the hubs it uses.
- 3 Its applications**
Its company details and live application status.
- 4 Its own pages**
Network, service levels, requests and diagnostics. Nothing else is reachable.

Why it matters

Customers can answer their own questions, which means fewer tickets and more trust.

Customer self-service requests

Prairie Grain Logistics

your managed network service · site cust2

Your network

Service levels

Requests

Diagnostics

Derek Olsen (Prairie Grain Logistics)

prairie · customer · cust2 · password · sign out · v0.21.0 · [dark](#)

1

Ask for a change

your request goes to our operators as a change request; you see here when it is approved and done

Applications (what you subscribe to)

☒ APP-1001 Unified Communications (VoIP)

☒ APP-1002 Email & Collaboration

☒ APP-1003 ERP — Finance & Procurement

☒ APP-1004 Point of Sale Backend

☐ APP-1005 Electronic Health Records

☒ APP-1006 Fleet Tracking & Telematics

☒ APP-1007 Inventory & Warehouse Management

☐ APP-1008 Secure File Transfer

☐ APP-1009 Video Surveillance (NVR)

☒ APP-1010 Identity & Single Sign-On

Preferred hub

hub-central (Central)

Second provider

☐ a second, backup circuit (dual-homed)

Why (optional)

Send the request

2

Our requests

Request	What	Asked	Status
CR-0012	modify cust2: preferred hub none → hub-central; applications APP-1001, APP-1002, APP-1003, APP-1006, APP-1007, APP-1010 → APP-1001, APP-1002, APP-1003, APP-1004, APP-1006, APP-1007, APP-1010 why: new point-of-sale rollout in our Chicago office	Sep 27, 03:13:06 PM · prairie	done approved for the POS rollout

- 1

Ask for a change

Applications, the preferred hub or a second provider, with a reason.
- 2

Track it

Each request becomes a change request for the operators, and its progress is shown here.

Why it matters
Customers can request changes without email, and the request still goes through staff approval.

Self-service diagnostics

Prairie Grain Logistics

your managed network service · site cust2

Your network

Service levels

Requests

Diagnostics

Derek Olsen (Prairie Grain Logistics) prairie · customer · cust2 · password · sign out · v0.21.0 · [dark](#)

1

Diagnostics

tests run now, from your own LAN host, to your applications and to our hubs

2

Test my applications

Test the hubs

APP-1001 Unified Communications (VoIP) at hub-east

Trace the path

Application	At	Answers	Round trip	Loss	HTTPS
APP-1001 Unified Communications (VoIP)	hub-east 192.168.71.10	yes	3.0 ms	0 %	not an HTTPS service
APP-1001 Unified Communications (VoIP)	hub-central 192.168.72.10	yes	3.4 ms	0 %	not an HTTPS service
APP-1001 Unified Communications (VoIP)	hub-west 192.168.73.10	yes	2.4 ms	0 %	not an HTTPS service
APP-1002 Email & Collaboration	hub-east 192.168.71.11	yes	2.7 ms	0 %	connects
APP-1002 Email & Collaboration	hub-west 192.168.73.11	yes	2.2 ms	0 %	connects
APP-1003 ERP — Finance & Procurement	hub-central 192.168.72.11	yes	2.0 ms	0 %	connects
APP-1004 Point of Sale Backend	hub-central 192.168.72.12	yes	1.8 ms	0 %	connects
APP-1004 Point of Sale Backend	hub-west 192.168.73.12	yes	1.1 ms	0 %	connects
APP-1006 Fleet Tracking & Telematics	hub-central 192.168.72.13	yes	1.1 ms	0 %	connects
APP-1007 Inventory & Warehouse Management	hub-central 192.168.72.14	yes	1.4 ms	0 %	connects
APP-1007 Inventory & Warehouse Management	hub-west 192.168.73.13	yes	1.4 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-east 192.168.71.14	yes	2.5 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-central 192.168.72.16	yes	1.7 ms	0 %	connects
APP-1010 Identity & Single Sign-On	hub-west 192.168.73.15	yes	1.0 ms	0 %	connects

5:53:51 PM · five pings to each, from your LAN host

- 1

Tests from its LAN host

The customer tests its applications, the hubs, or traces a path.
- 2

Every application, every hub

Round-trip time, loss and HTTP status, run on demand.

Why it matters

The first-line diagnosis is done by the customer in seconds, with no ticket needed.

Lab tools and accounts

C8000v DMVPN Portal

three hubs · Catalyst 8000v and VyOS customers · two VyOS providers · Network-as-Code · Nautobot

The cloudNetwork mapSLAResilienceProvisionJobsLab ToolsAPI

Lab administratoradmin · admin · password · sign out · v0.21.0 · 5 customers · 1 shortcuts · healthy · [C_dark](#)

1

Users

who may sign in, and what each role may do — viewer reads, operator changes, approver approves, admin all, customer its own service

Account	Name	Roles	Customer	State	
admin	Lab administrator	admin		active lab-default password	set passworddisable
approver	Change approver	approver		active lab-default password	set passworddisabledelete
operator	Network operator	operator		active lab-default password	set passworddisabledelete
prairie	Derek Olsen (Prairie Grain Logistics)	customer	cust2	active	set passworddisabledelete
viewer	Read-only viewer	viewer		active lab-default password	set passworddisabledelete

Username

Full name

Role

Password (8+)

viewer

Add the account

How to get in

Lab host: ssh dcantor@192.168.50.231 then cd /home/dcantor/c8000v-dmvpn-lab management network 10.5.0.0/24 (c8d-oob, host 10.5.0.1)

The management addresses (10.5.0.0/24) and the consoles live on the lab host (bridge c8d-oob, host 10.5.0.1): log in to the lab host first, then ssh from there or run ./lab.sh in /home/dcantor/c8000v-dmvpn-lab.

2

Tools

everything the lab uses, with the login each one takes

LAB

Provisioning portal

the cloud, the Network map, provisioning jobs; customers sign in to their own view

login: an account: admin / operator / approver / viewer, lab defaults in webapp/users.seed.json

http://192.168.50.231:8094

Portal API (Swagger)

every endpoint the portal offers; /metrics and /api/sd for Prometheus stay open

login: the portal's session (sign in first)

http://192.168.50.231:8094/docs

Lab hub

every lab on this host, VM power, host CPU / memory

login: none

http://192.168.50.231:8088

GitHub repository

the code: lab.sh, the renderer, NAC, tests, Nautobot, the portal

login: public

https://github.com/dcantor/c8000v-dmvpn-lab

SOURCE OF TRUTH

Nautobot

the shared NMS (cat9000v lab); API token: ./lab.sh nautobot token

Nautobot: this lab's devices

every VM of the lab, under its c8d- name

Nautobot: customer companies

a tenant per customer, with its details and subscriptions

Nautobot: applications (Virtual Servers)

one per application per hosting hub, with its VIP

- 1

Accounts and roles

Admins add, disable and delete accounts and set passwords. Accounts still on their lab-default password are flagged.
- 2

Every tool

Nautobot, the API, the lab hub and the repository, each with how to sign in.

Why it matters
Access is managed in one place, with visible warnings when the defaults have not been changed.

Everything is an API

1

c8000v-dmvpn-lab Provisioning Portal API 1.0 OAS 3.1

/openapi.json

REST API behind the C8000v DMVPN lab's portal. Every change goes **lab.conf** → **rendered configuration** → **VMs** → **Network-as-Code** → **provider** → **Nautobot** → **verification** → **tests**; runs are asynchronous (**POST** **/api/runs** , poll **GET** **/api/runs/{id}**). UI: /

2

monitoring Prometheus: /metrics and /api/sd, scraped by the NMS.

GET **/metrics** Prometheus metrics: VMs, the cloud per router, the provider, hosts, runs

GET **/api/sd** Prometheus HTTP service discovery: the portal, the provider's exporters, the hosts

state The cloud, the model and the live state of every router.

GET **/api/version** The lab's version (VERSION; CHANGELOG.md has what changed)

GET **/api/state** The cloud: the model, and what every router in it is doing

GET **/api/config-history** Jobs that changed configurations, newest first (optionally one router's)

GET **/api/config-history/{job}/{router}** A router's configuration before or after a job

GET **/api/drift** The latest configuration drift report

POST **/api/drift/check** Run the drift check now, outside the run queue, and return the report

GET **/api/query/{node}** One read-only 'show' command on one C8000v

GET **/api/customers/{name}/sla** A customer's service levels over a window, from VictoriaMetrics

GET **/api/hosts/{host}/ping** Ping one LAN host from another, over the overlay

GET **/api/hosts/{host}/traceroute** Trace the path from one LAN host to another, hop by hop

1

REST API

Every screen in the portal is built on the same API, documented with Swagger.

2

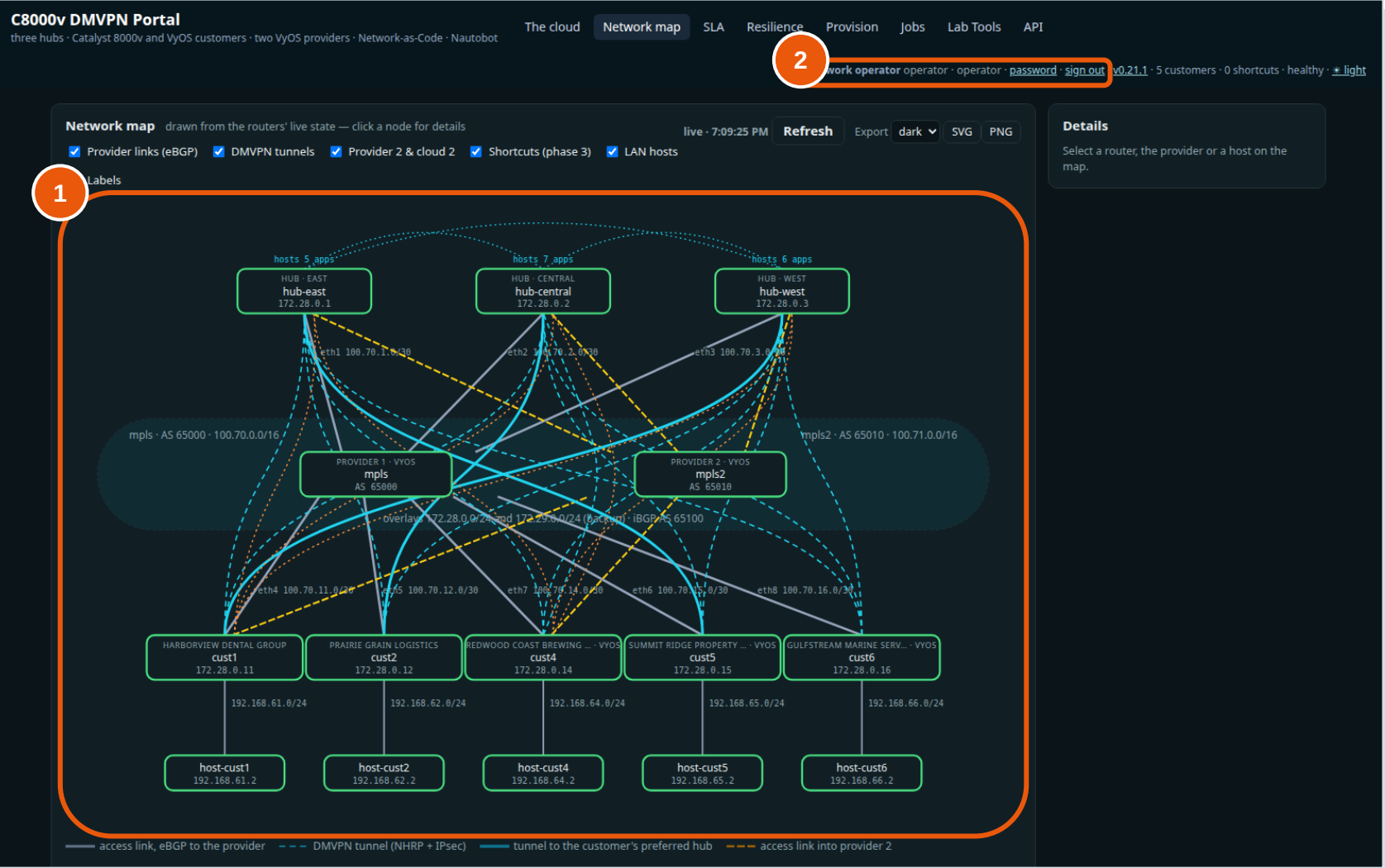
By area

Monitoring, state, customers, jobs, changes, SLA, and more.

Why it matters

Other tools and pipelines can drive the lab the way the portal does.

Light or dark, and roles everywhere



1 **Dark mode**
Every page, including the live map.

2 **An operator**
It sees and changes the lab, but admin tasks such as key rotation and accounts are hidden.

Why it matters
The portal fits the user, and each role only sees the actions it may take.



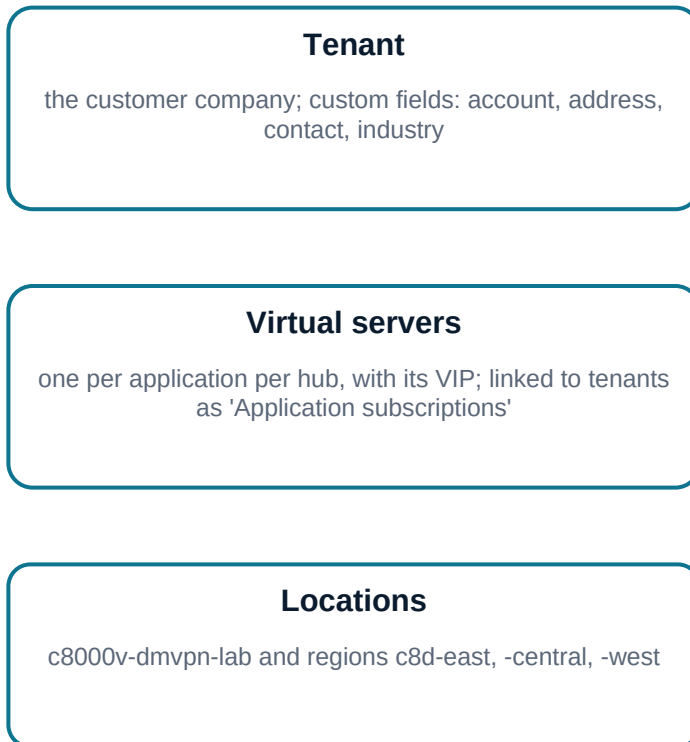
Source of truth: Nautobot

Where the customers, their applications and the VPN are modelled.

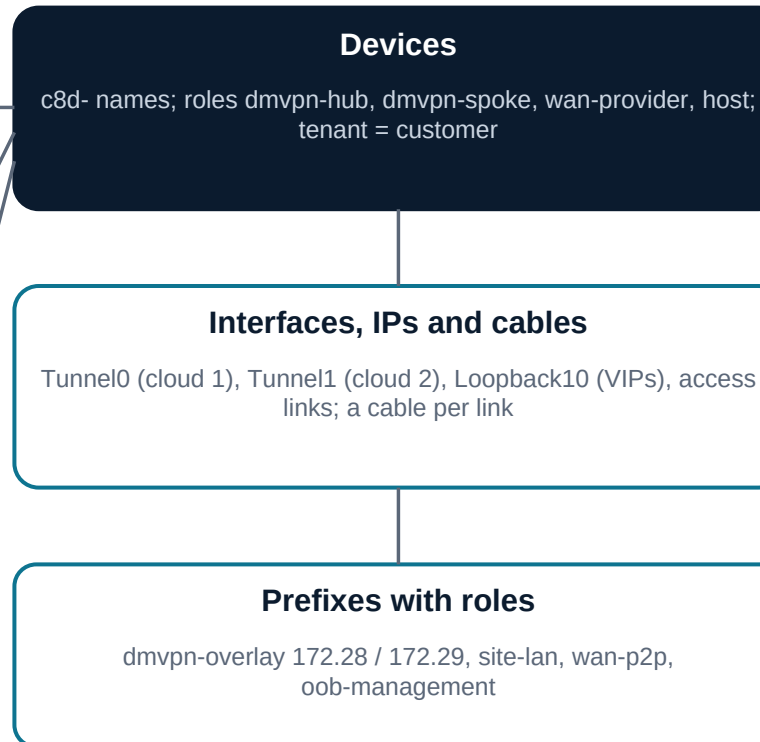
Where the customers and the VPN live in Nautobot

lab.sh nautobot seed writes the whole lab into the shared Nautobot. The renderer can build every configuration from Nautobot alone, and a check proves the result is byte for byte the same as from lab.conf.

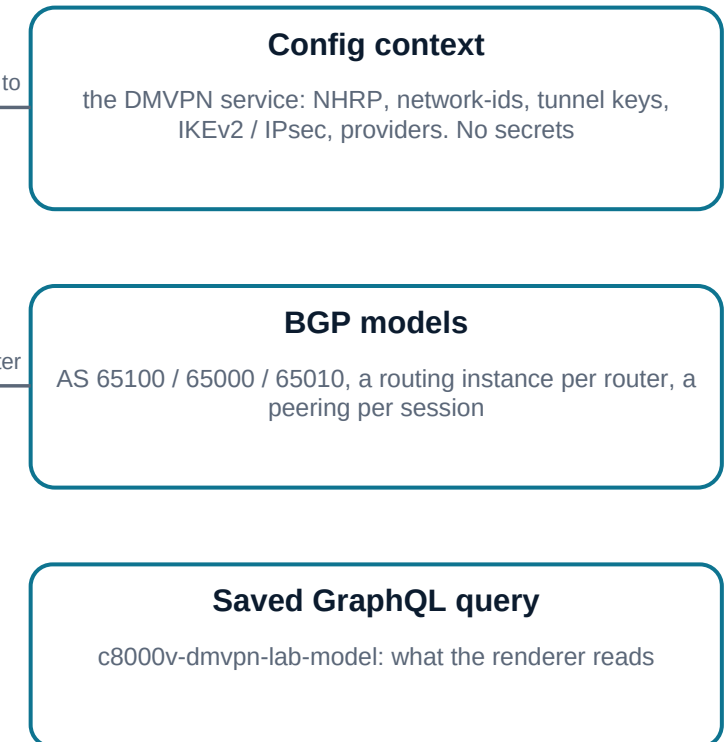
THE CUSTOMER



THE NETWORK



THE VPN SERVICE



Deliberately not in Nautobot: the pre-shared key and the NHRP secret. They stay in secrets/ on the lab host.

Nautobot: every device of the lab

cat9000v lab NMS

admin

+ Add Device

Name	Status	Tenant	Role	Type	Location	Rack	IP Address
c8d-cust1	Active	Harborview Dental Group	dmvpn-spoke	Cisco C8000V	c8d-east	—	10.5.0.21/24
c8d-cust2	Active	Prairie Grain Logistics	dmvpn-spoke	Cisco C8000V	c8d-central	—	10.5.0.22/24
c8d-cust4	Active	Redwood Coast Brewing Co.	dmvpn-spoke	VyOS VyOS	c8d-east	—	10.5.0.24/24
c8d-cust5	Active	Summit Ridge Property Management	dmvpn-spoke	VyOS VyOS	c8d-central	—	10.5.0.25/24
c8d-cust6	Active	Gulfstream Marine Services	dmvpn-spoke	VyOS VyOS	c8d-west	—	10.5.0.26/24
c8d-host-cust1	Active	Harborview Dental Group	host	Alpine Linux Alpine VM	c8d-east	—	10.5.0.41/24
c8d-host-cust2	Active	Prairie Grain Logistics	host	Alpine Linux Alpine VM	c8d-central	—	10.5.0.42/24
c8d-host-cust4	Active	Redwood Coast Brewing Co.	host	Alpine Linux Alpine VM	c8d-east	—	10.5.0.44/24
c8d-host-cust5	Active	Summit Ridge Property Management	host	Alpine Linux Alpine VM	c8d-central	—	10.5.0.45/24
c8d-host-cust6	Active	Gulfstream Marine Services	host	Alpine Linux Alpine VM	c8d-west	—	10.5.0.46/24
c8d-hub-central	Active	—	dmvpn-hub	Cisco C8000V	c8d-central	—	10.5.0.12/24
c8d-hub-east	Active	—	dmvpn-hub	Cisco C8000V	c8d-east	—	10.5.0.11/24
c8d-hub-west	Active	—	dmvpn-hub	Cisco C8000V	c8d-west	—	10.5.0.13/24
c8d-mpls	Active	—	wan-provider	VyOS VyOS	c8000v-dmvpn-lab	—	10.5.0.31/24
c8d-mpls2	Active	—	wan-provider	VyOS VyOS	c8000v-dmvpn-lab	—	10.5.0.32/24

Edit Selected + Add Components

Showing 1-15 of 15 1000 per page

2e12edd1d0ed (v3.2.4) 2026-09-28 00:18

- The lab's devices**
Routers, providers and LAN hosts, under their c8d- names in the shared Nautobot.
- Tenant**
Each customer's router and host belong to its company.
- Role**
dmvpn-hub, dmvpn-spoke, wan-provider, host.
- Location**
Its region: c8d-east, -central or -west.

Why it matters

One inventory for every lab on the host; this lab's part is easy to find and to filter.

Nautobot: a customer company as a tenant

The screenshot displays the Nautobot web interface for a tenant named 'Prairie Grain Logistics'. The interface is divided into several sections, with four key areas highlighted by orange boxes and numbered annotations:

- 1 Tenant:** The top section of the tenant record, showing the tenant name 'Prairie Grain Logistics', the tenant group 'c8000v-dmvpn-lab customers', and a description 'Transportation & logistics - customer site cust2 (Central)'.
- 2 Custom fields:** A section below the tenant information containing a table of custom fields. The table has two columns: 'Field Name' and 'Value'. The fields are categorized under 'ACME' and 'Customer'.
- 3 Subscribed applications:** A section at the bottom of the tenant record showing a list of subscribed applications. The list includes 'APP-1001 Unified Communications (VoIP) @ hub-central', 'APP-1001 Unified Communications (VoIP) @ hub-east', 'APP-1001 Unified Communications (VoIP) @ hub-west', and '11 other Virtual Servers'.
- 4 Its devices:** A section on the right side of the tenant record showing a grid of device counts for various categories. The categories and their counts are: Circuits (0), Clusters (0), Controllers (0), Controller Managed Device Groups (0), Devices (2), Dynamic Groups (0), IP Addresses (0), IP Address Ranges (0), Locations (0), Namespaces (0), Prefixes (0), Racks (0), Rack Reservations (0), Virtual Machines (0), VLANs (0), and VRFs (0).

- 1 Tenant**
Prairie Grain Logistics, in the c8000v-dmvpn-lab customers group.
- 2 Custom fields**
Account, address, contact, email, industry, phone.
- 3 Subscribed applications**
A relationship to the virtual servers it uses, at every hub.
- 4 Its devices**
Its router and its LAN host.

Why it matters

The customer record the portal edits is the one Nautobot holds: no second customer list to keep in sync.

Nautobot: applications as virtual servers

The screenshot shows the Nautobot interface for managing virtual servers. The left sidebar contains navigation links for various system components. The main panel displays a table of virtual servers, each representing an application instance across different hubs. The table includes columns for Name, IP, Enabled status, Load balancer type, Port, Protocol, and Load balancer pool. An orange box labeled '1' highlights the left sidebar, and another orange box labeled '2' highlights the 'IP' column in the table.

Name	IP	Enabled	Load balancer type	Port	Protocol	Load balancer pool
APP-1001 Unified Communications (VoIP) @ hub-central	192.168.72.10/32	✓	Layer 4	5061	TCP	—
APP-1001 Unified Communications (VoIP) @ hub-east	192.168.71.10/32	✓	Layer 4	5061	TCP	—
APP-1001 Unified Communications (VoIP) @ hub-west	192.168.73.10/32	✓	Layer 4	5061	TCP	—
APP-1002 Email & Collaboration @ hub-east	192.168.71.11/32	✓	Layer 7	443	HTTPS	—
APP-1002 Email & Collaboration @ hub-west	192.168.73.11/32	✓	Layer 7	443	HTTPS	—
APP-1003 ERP — Finance & Procurement @ hub-central	192.168.72.11/32	✓	Layer 7	443	HTTPS	—
APP-1004 Point of Sale Backend @ hub-central	192.168.72.12/32	✓	Layer 7	443	HTTPS	—
APP-1004 Point of Sale Backend @ hub-west	192.168.73.12/32	✓	Layer 7	443	HTTPS	—
APP-1005 Electronic Health Records @ hub-east	192.168.71.12/32	✓	Layer 7	443	HTTPS	—
APP-1006 Fleet Tracking & Telematics @ hub-central	192.168.72.13/32	✓	Layer 7	443	HTTPS	—
APP-1007 Inventory & Warehouse Management @ hub-central	192.168.72.14/32	✓	Layer 7	443	HTTPS	—
APP-1007 Inventory & Warehouse Management @ hub-west	192.168.73.13/32	✓	Layer 7	443	HTTPS	—
APP-1008 Secure File Transfer @ hub-central	192.168.72.15/32	✓	Layer 4	22	TCP	—
APP-1008 Secure File Transfer @ hub-east	192.168.71.13/32	✓	Layer 4	22	TCP	—
APP-1009 Video Surveillance (NVR) @ hub-west	192.168.73.14/32	✓	Layer 4	554	TCP	—
APP-1010 Identity & Single Sign-On @ hub-central	192.168.72.16/32	✓	Layer 7	443	HTTPS	—
APP-1010 Identity & Single Sign-On @ hub-east	192.168.71.14/32	✓	Layer 7	443	HTTPS	—
APP-1010 Identity & Single Sign-On @ hub-west	192.168.73.15/32	✓	Layer 7	443	HTTPS	—

Showing 1-18 of 18 | 1000 per page

1 One per application per hub
APP-1001 to APP-1010, each at the hubs that host it.

2 Its VIP
An address on the hub's Loopback10, with port and protocol; the portal's application checks test these.

Why it matters

Applications and who subscribes to them are modelled, not just the routers.

Nautobot: a hub

>>> nautobot

FAVORITES

DEVICES

IPAM

ORGANIZATION

CIRCUITS

VPN

ROUTING

POWER

WIRELESS

SECRETS

LOAD BALANCERS

VIRTUALIZATION

CLOUD

APPROVALS

EXTENSIBILITY

GOLDEN CONFIG

JOBS

APPS

cat9000v lab NMS

>>> c8d-hub-east

in: Devices × Press Ctrl+K to search

admin

Device

Advanced

Contacts 0

Dynamic Groups 1

Object Metadata 0

Data Compliance 0

Interfaces 8

Status

LLDP Neighbors

Configuration

Sept. 26, 2026 10:53 p.m. | 1 day, 1 hour ago

+ Add Components

Edit Device

1

DEVICE

Location

c8000v-dmvpn-lab (Site)

c8d-east (Region)

Rack

—

Position (U)

—

Tenant

—

Device Family

—

Device Type

Cisco C8000V (1U)

Serial Number

—

Asset Tag

—

MANAGEMENT

Role

dmvpn-hub

Platform

cisco_xe

Status

Active

Primary IPv4

10.5.0.11/24 (Global)

Primary IPv6

—

Secrets Group

—

Device Redundancy Group

—

2

ASSIGNED VRFS 0

VRFS

— No VRF-device assignments found —

CLUSTERS 0

CLUSTERS

— No clusters found —

+ Add cluster

SERVICES 0

SERVICES

— No services found —

+ Add service

IMAGES 0

IMAGES

— No image attachments found —

+ Add image attachment

3

AUTONOMOUS SYSTEMS 1

AUTONOMOUS SYSTEMS

ASN

65100

- 1

The device
Location, role dmvpn-hub, platform, management address, autonomous system.
- 2

Interfaces
Its tunnels, loopbacks and access links.
- 3

Autonomous system
AS 65100, from the BGP models; the Config Context tab shows the DMVPN service data that applies to it.

Why it matters
Everything needed to build this hub's configuration is on this page or one click away.

Nautobot: the VPN on the hub's interfaces

The screenshot displays the Nautobot web interface for the 'c8d-hub-east' device. The 'INTERFACES' tab is selected, showing a list of interfaces. Four interfaces are highlighted with orange boxes and numbered 1 through 4:

- 1 Tunnel0**: Cloud 1: mGRE, phase 3, sourced from GigabitEthernet2.
- 2 Tunnel1**: Cloud 2, the backup: sourced from GigabitEthernet4 into mpls2.
- 3 Loopback10**: The hub LAN, carrying the application VIPs.
- 4 Access link**: Cabled to the provider's port.

The interface table includes columns for Name, Status, Role, Label, Module, Enabled, Type, Port type, Speed, Parent, Breakout position, LAG, MTU, VRF, 802.1Q Mode, and Description. The highlighted interfaces are:

- GigabitEthernet2**: Active, 1000BASE-T (1GE), OOB management.
- GigabitEthernet4**: Active, 1000BASE-T (1GE), provider: mpls2 eth1.
- Loopback10**: Active, Virtual, hub LAN.
- Tunnel0**: Active, Virtual, DMVPN hub (mGRE, phase 3), sourced from GigabitEthernet2 100.70.1.2.
- Tunnel1**: Active, Virtual, DMVPN hub, cloud 2 (mGRE, phase 3), sourced from GigabitEthernet4 100.71.1.1.

- Tunnel0**
Cloud 1: mGRE, phase 3, sourced from GigabitEthernet2.
- Tunnel1**
Cloud 2, the backup: sourced from GigabitEthernet4 into mpls2.
- Loopback10**
The hub LAN, carrying the application VIPs.
- Access link**
Cabled to the provider's port.

Why it matters

The two DMVPN clouds are modelled as real interfaces with their sources, addresses and cables.

Nautobot: the addressing plan

cat9000v lab NMS

Prefixes

in: Prefixes Press Ctrl+K to search

admin

Max Length + Add Prefix

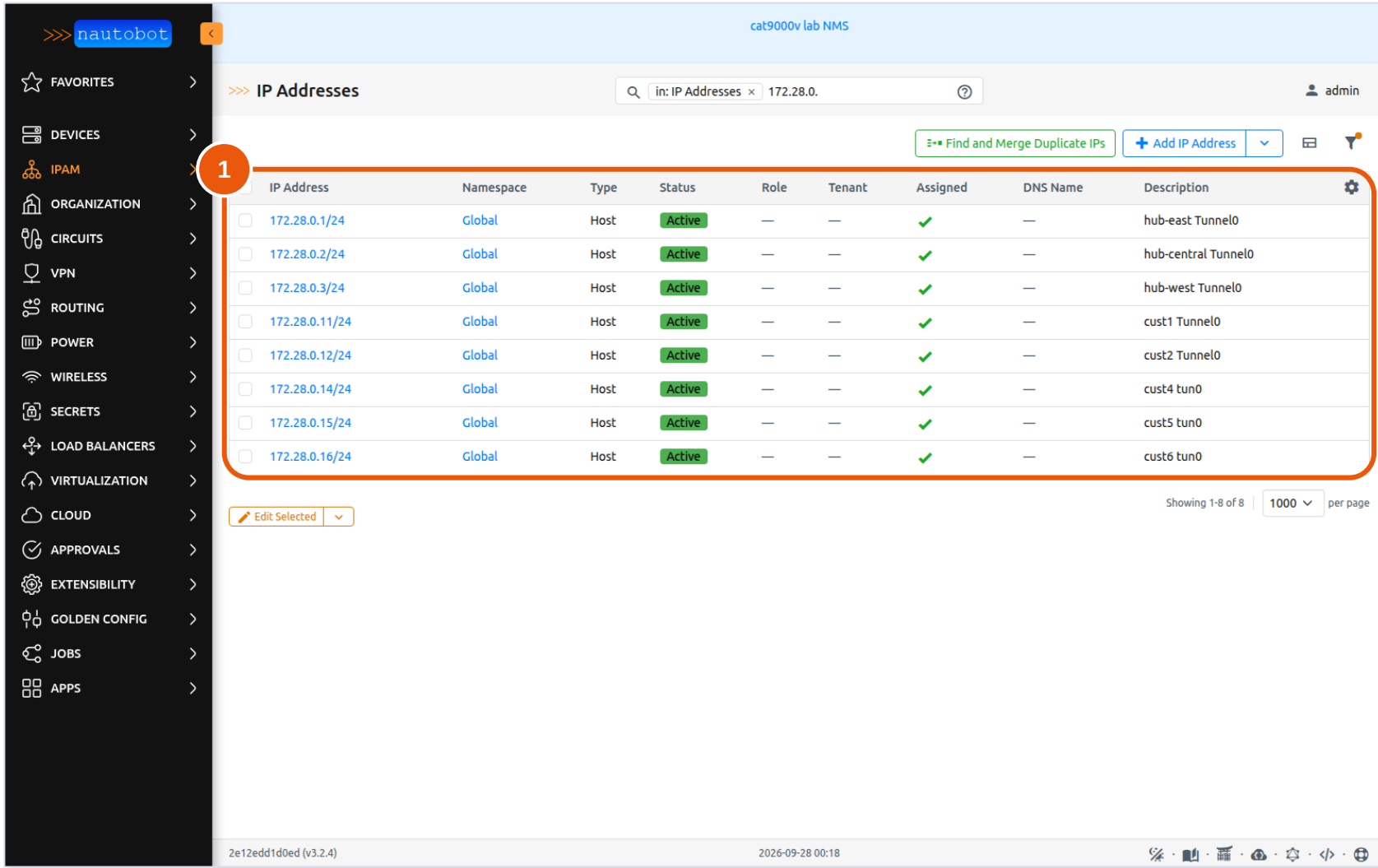
Prefix	Namespace	Type	Status	Descendants	VRFs	Tenant	Locations	VL	Role	Description
192.168.73.0/24	Global	Network	Active	—	—	—	c8d-west	—	site-lan	hub-west LAN (Loopback10)
192.168.72.0/24	Global	Network	Active	—	—	—	c8d-central	—	site-lan	hub-central LAN (Loopback10)
192.168.71.0/24	Global	Network	Active	—	—	—	c8d-east	—	site-lan	hub-east LAN (Loopback10)
192.168.66.0/24	Global	Network	Active	—	—	—	c8d-west	—	site-lan	cust6 eth3 <=> host-cust6 eth1
192.168.65.0/24	Global	Network	Active	—	—	—	c8d-central	—	site-lan	cust5 eth3 <=> host-cust5 eth1
192.168.64.0/24	Global	Network	Active	—	—	—	c8d-east	—	site-lan	cust4 eth3 <=> host-cust4 eth1
192.168.62.0/24	Global	Network	Active	—	—	—	c8d-central	—	site-lan	cust2 GigabitEthernet3 <=> host-cust2 eth1
192.168.61.0/24	Global	Network	Active	—	—	—	c8d-east	—	site-lan	cust1 GigabitEthernet3 <=> host-cust1 eth1
172.29.0.0/24	Global	Network	Active	—	—	—	c8000v-dmvpn-lab	—	dmvpn-overlay	the second DMVPN cloud (backup): Tunnel1 on the hubs and the dual-homed customers, network-id 2, tunnel key 200
172.28.0.0/24	Global	Network	Active	—	—	—	c8000v-dmvpn-lab	—	dmvpn-overlay	the DMVPN cloud: Tunnel0 on every router, network-id 1, tunnel key 100
100.71.14.0/30	Global	Network	Active	—	—	—	c8d-east	—	wan-p2p	mpls2 eth4 <=> cust4 eth4
100.71.11.0/30	Global	Network	Active	—	—	—	c8d-east	—	wan-p2p	mpls2 eth5 <=> cust1 GigabitEthernet4
100.71.3.0/30	Global	Network	Active	—	—	—	c8d-west	—	wan-p2p	mpls2 eth3 <=> hub-west GigabitEthernet4
100.71.2.0/30	Global	Network	Active	—	—	—	c8d-central	—	wan-p2p	mpls2 eth2 <=> hub-central GigabitEthernet4
100.71.1.0/30	Global	Network	Active	—	—	—	c8d-east	—	wan-p2p	mpls2 eth1 <=> hub-east GigabitEthernet4
100.71.0.0/16	Global	Container	Active	5	—	—	c8000v-dmvpn-lab	—	wan-p2p	the second provider's access links (AS 65010): one /30 per hub and dual-homed customer, carried by eBGP
100.70.16.0/30	Global	Network	Active	—	—	—	c8d-west	—	wan-p2p	mpls eth8 <=> cust6 eth2

- Cloud 1**
172.28.0.0/24, role dmvpn-overlay:
network-id 1, tunnel key 100.
- Cloud 2**
172.29.0.0/24: network-id 2, tunnel key 200.
- Site LANs**
Each customer's LAN, role site-lan.
- Roles**
site-lan, dmvpn-overlay, wan-p2p,
oob-management.

Why it matters

Every prefix says what it is for, so the allocator and the renderer agree.

Nautobot: tunnel addresses



The screenshot shows the Nautobot IPAM interface. The left sidebar contains navigation links for FAVORITES, DEVICES, IPAM (highlighted), ORGANIZATION, CIRCUITS, VPN, ROUTING, POWER, WIRELESS, SECRETS, LOAD BALANCERS, VIRTUALIZATION, CLOUD, APPROVALS, EXTENSIBILITY, GOLDEN CONFIG, JOBS, and APPS. The main content area is titled 'IP Addresses' and shows a search bar with 'in: IP Addresses' and '172.28.0.'. Below the search bar are buttons for 'Find and Merge Duplicate IPs', 'Add IP Address', and a filter icon. A table lists IP addresses, with a red box highlighting the first 8 rows. A red circle with the number 1 points to the first row of the table.

IP Address	Namespace	Type	Status	Role	Tenant	Assigned	DNS Name	Description
172.28.0.1/24	Global	Host	Active	—	—	✓	—	hub-east Tunnel0
172.28.0.2/24	Global	Host	Active	—	—	✓	—	hub-central Tunnel0
172.28.0.3/24	Global	Host	Active	—	—	✓	—	hub-west Tunnel0
172.28.0.11/24	Global	Host	Active	—	—	✓	—	cust1 Tunnel0
172.28.0.12/24	Global	Host	Active	—	—	✓	—	cust2 Tunnel0
172.28.0.14/24	Global	Host	Active	—	—	✓	—	cust4 tun0
172.28.0.15/24	Global	Host	Active	—	—	✓	—	cust5 tun0
172.28.0.16/24	Global	Host	Active	—	—	✓	—	cust6 tun0

Showing 1-8 of 8 | 1000 per page

- Tunnel0 addresses**
Each router's address in cloud 1, assigned to its Tunnel0: hubs .1 to .3, customers from their index.

Why it matters

NHRP registrations and BGP neighbours come straight from these addresses.

Nautobot: every BGP session

cat9000v lab NMS

in: Peerings x c8d

+ Add BGP Peering

Peering	Endpoint	Endpoint	Status	Dynamic Groups
☐ c8d-hub-west 172.29.0.3/24 (AS 65100) ↔ c8d-hub-central 172.29.0.2/24 (AS 65100)	172.29.0.3/24	172.29.0.2/24	Active	—
☐ c8d-hub-east 172.28.0.1/24 (AS 65100) ↔ c8d-cust2 172.28.0.12/24 (AS 65100)	172.28.0.1/24	172.28.0.12/24	Active	—
☐ c8d-hub-east 172.28.0.1/24 (AS 65100) ↔ c8d-cust4 172.28.0.14/24 (AS 65100)	172.28.0.1/24	172.28.0.14/24	Active	—
☐ c8d-mpls 100.70.3.1/30 (AS 65000) ↔ c8d-hub-west 100.70.3.2/30 (AS 65100)	100.70.3.1/30	100.70.3.2/30	Active	—
☐ c8d-hub-east 100.71.1.1/30 (AS 65100) ↔ c8d-mpls2 100.71.1.2/30 (AS 65010)	100.71.1.1/30	100.71.1.2/30	Active	—
☐ c8d-hub-west 172.29.0.3/24 (AS 65100) ↔ c8d-hub-east 172.29.0.1/24 (AS 65100)	172.29.0.3/24	172.29.0.1/24	Active	—
☐ c8d-cust2 172.28.0.12/24 (AS 65100) ↔ c8d-hub-central 172.28.0.2/24 (AS 65100)	172.28.0.12/24	172.28.0.2/24	Active	—
☐ c8d-hub-east 172.28.0.1/24 (AS 65100) ↔ c8d-cust1 172.28.0.11/24 (AS 65100)	172.28.0.1/24	172.28.0.11/24	Active	—
☐ c8d-cust6 172.28.0.16/24 (AS 65100) ↔ c8d-hub-east 172.28.0.1/24 (AS 65100)	172.28.0.16/24	172.28.0.1/24	Active	—
☐ c8d-cust5 100.70.15.2/30 (AS 65100) ↔ c8d-mpls 100.70.15.1/30 (AS 65000)	100.70.15.2/30	100.70.15.1/30	Active	—
☐ c8d-mpls2 100.71.3.2/30 (AS 65010) ↔ c8d-hub-west 100.71.3.1/30 (AS 65100)	100.71.3.2/30	100.71.3.1/30	Active	—
☐ c8d-hub-central 172.28.0.2/24 (AS 65100) ↔ c8d-cust5 172.28.0.15/24 (AS 65100)	172.28.0.2/24	172.28.0.15/24	Active	—
☐ c8d-cust4 172.28.0.14/24 (AS 65100) ↔ c8d-hub-central 172.28.0.2/24 (AS 65100)	172.28.0.14/24	172.28.0.2/24	Active	—
☐ c8d-hub-east 172.28.0.1/24 (AS 65100) ↔ c8d-hub-west 172.28.0.3/24 (AS 65100)	172.28.0.1/24	172.28.0.3/24	Active	—
☐ c8d-cust4 172.29.0.14/24 (AS 65100) ↔ c8d-hub-east 172.29.0.1/24 (AS 65100)	172.29.0.14/24	172.29.0.1/24	Active	—
☐ c8d-mpls2 100.71.11.1/30 (AS 65010) ↔ c8d-cust1 100.71.11.2/30 (AS 65100)	100.71.11.1/30	100.71.11.2/30	Active	—
☐ c8d-cust4 100.70.14.2/30 (AS 65100) ↔ c8d-mpls 100.70.14.1/30 (AS 65000)	100.70.14.2/30	100.70.14.1/30	Active	—
☐ c8d-mpls 100.70.12.1/30 (AS 65000) ↔ c8d-cust2 100.70.12.2/30 (AS 65100)	100.70.12.1/30	100.70.12.2/30	Active	—
☐ c8d-hub-central 172.29.0.2/24 (AS 65100) ↔ c8d-cust1 172.29.0.11/24 (AS 65100)	172.29.0.2/24	172.29.0.11/24	Active	—
☐ c8d-hub-west 172.28.0.3/24 (AS 65100) ↔ c8d-cust5 172.28.0.15/24 (AS 65100)	172.28.0.3/24	172.28.0.15/24	Active	—
☐ c8d-mpls 100.70.16.1/30 (AS 65000) ↔ c8d-cust6 100.70.16.2/30 (AS 65100)	100.70.16.1/30	100.70.16.2/30	Active	—

1

Peerings

Hub ↔ hub, hub (route reflector) ↔ customer, and site ↔ provider, over both clouds and both providers.

Why it matters

The routing design is data: a missing or extra session shows up as a difference, not a surprise.

Nautobot: BGP routing instances

1

Routing instance	Device	Autonomous system	Router ID	Status
☐ c8d-mpls - AS 65000	c8d-mpls	AS 65000	10.255.5.254/32	Active
☐ c8d-cust5 - AS 65100	c8d-cust5	AS 65100	10.255.5.15/32	Active
☐ c8d-cust6 - AS 65100	c8d-cust6	AS 65100	10.255.5.16/32	Active
☐ c8d-mpls2 - AS 65010	c8d-mpls2	AS 65010	10.255.5.253/32	Active
☐ c8d-hub-east - AS 65100	c8d-hub-east	AS 65100	10.255.5.1/32	Active
☐ c8d-hub-central - AS 65100	c8d-hub-central	AS 65100	10.255.5.2/32	Active
☐ c8d-hub-west - AS 65100	c8d-hub-west	AS 65100	10.255.5.3/32	Active
☐ c8d-cust4 - AS 65100	c8d-cust4	AS 65100	10.255.5.14/32	Active
☐ c8d-cust1 - AS 65100	c8d-cust1	AS 65100	10.255.5.11/32	Active
☐ c8d-cust2 - AS 65100	c8d-cust2	AS 65100	10.255.5.12/32	Active

Edit Selected

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1

Routing instances

One per router: its autonomous system (65100 sites, 65000 and 65010 providers) and router-id.

Why it matters

AS numbers and router-ids are allocated once and read from here.

Nautobot: the DMVPN service as a config context



1

```
{
  "dmvpn": {
    "as": 65100,
    "cloud2": {
      "network_id": 2,
      "overlay": "172.29.0.0/24",
      "tunnel_key": 200
    },
    "holdtime": 60,
    "holdtime_bgp": 9,
    "hubs": [
      "hub-east",
      "hub-central",
      "hub-west"
    ],
    "keepalive": 3,
    "mss": 1360,
    "mtu": 1400,
    "network_id": 1,
    "overlay": "172.28.0.0/24",
    "spokes": [
      "cust1",
      "cust2",
      "cust4",
      "cust5",
      "cust6"
    ],
    "tunnel_key": 100
  },
  "domain_name": "lab.local",
  "domain_prefix": "c8d-",
  "ipsec": {
    "ikev2_profile": "DMVPN-IKEV2",
    "ipsec_profile": "DMVPN-IPSEC",
    "key_exchange": "ikev2",
    "mode": "tunnel",
    "proposal": {
      "dh_group": 14,
      "encryption": "aes-cbc-256",
      "integrity": "sha256"
    }
  },
  "lab": "c8000v-dmvpn-lab",
  "mac_oil": "52:54:00:c9",
  "oob": {
    "gateway": "10.5.0.1",
```

- 1 **The service**
AS, both clouds (overlay, network-id, tunnel key), NHRP hold times, hubs and spokes, the IKEv2 and IPsec proposals, the providers. No pre-shared key.

Why it matters

The VPN service's settings are one reviewed document, applied to the whole lab.

Nautobot: the query the renderer reads

The screenshot shows the Nautobot web interface. On the left is a dark sidebar with navigation links: FAVORITES, DEVICES, IPAM, ORGANIZATION, CIRCUITS, VPN, ROUTING, POWER, WIRELESS, SECRETS, LOAD BALANCERS, VIRTUALIZATION, CLOUD, APPROVALS, EXTENSIBILITY, GOLDEN CONFIG, JOBS, and APPS. The top header is light blue with the Nautobot logo and the text 'cat9000v lab NMS'. Below the header, there's a search bar and a user profile 'admin'. The main content area has a tabbed interface. The 'GraphQL Query' tab is selected, showing a query editor. The query editor has a text area with the following GraphQL query:

```
query c8000v_dmvpn_lab_model {
  config_contexts(name: ["c8000v-dmvpn-lab"]) { name data }
  devices(location: ["c8000v-dmvpn-lab", "c8d-east", "c8d-central", "c8d-west"]) {
    name
    role { name }
    platform { name }
    location { name }
    primary_ip4 { address }
    interfaces {
      name type mgmt_only mac_address
      ip_addresses { address parent { prefix } }
      connected_interface { name device { name role { name } } ip_addresses {
        }
      }
    }
    bgp_routing_instances {
      autonomous_system { asn }
      router_id { address }
      extra_attributes
    }
  }
}
```

Below the query editor is a 'Variables' section with an empty object `{}`. To the right of the query editor is a 'RESPONSE' section with an 'Execute' button. An orange circle with the number '1' highlights the query editor area.

- Saved GraphQL query**
c8000v-dmvpn-lab-model: the config context, devices, interfaces, IPs and BGP in one call. The same renderer turns it into every configuration.

Why it matters

lab.sh nautobot render --check proves Nautobot and lab.conf produce identical configurations.

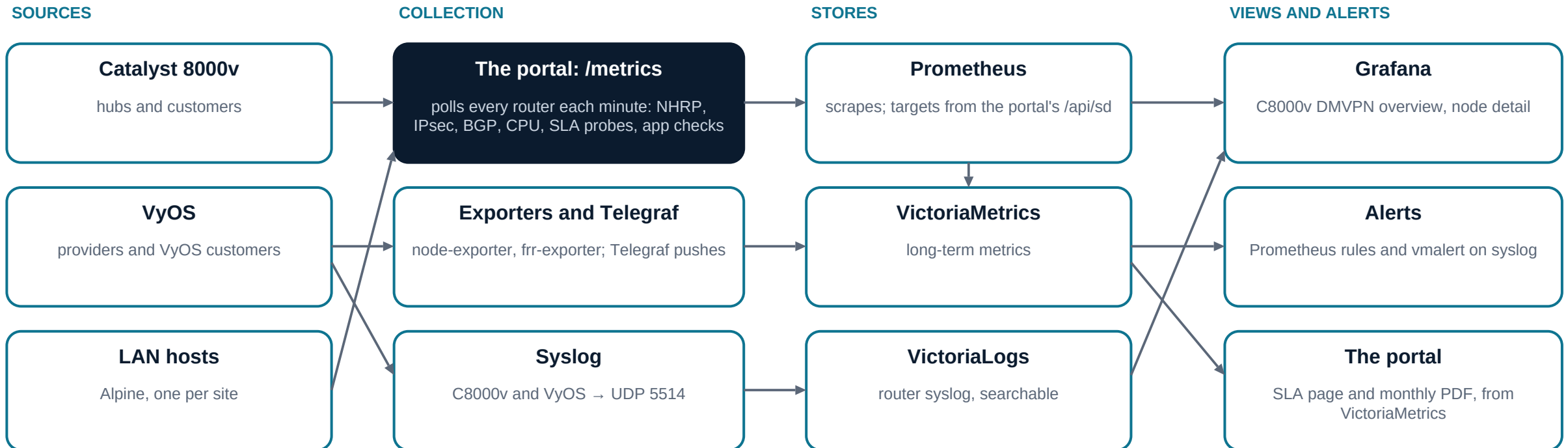


Monitoring

Prometheus, VictoriaMetrics, VictoriaLogs and Grafana: the lab over time.

How the lab is monitored

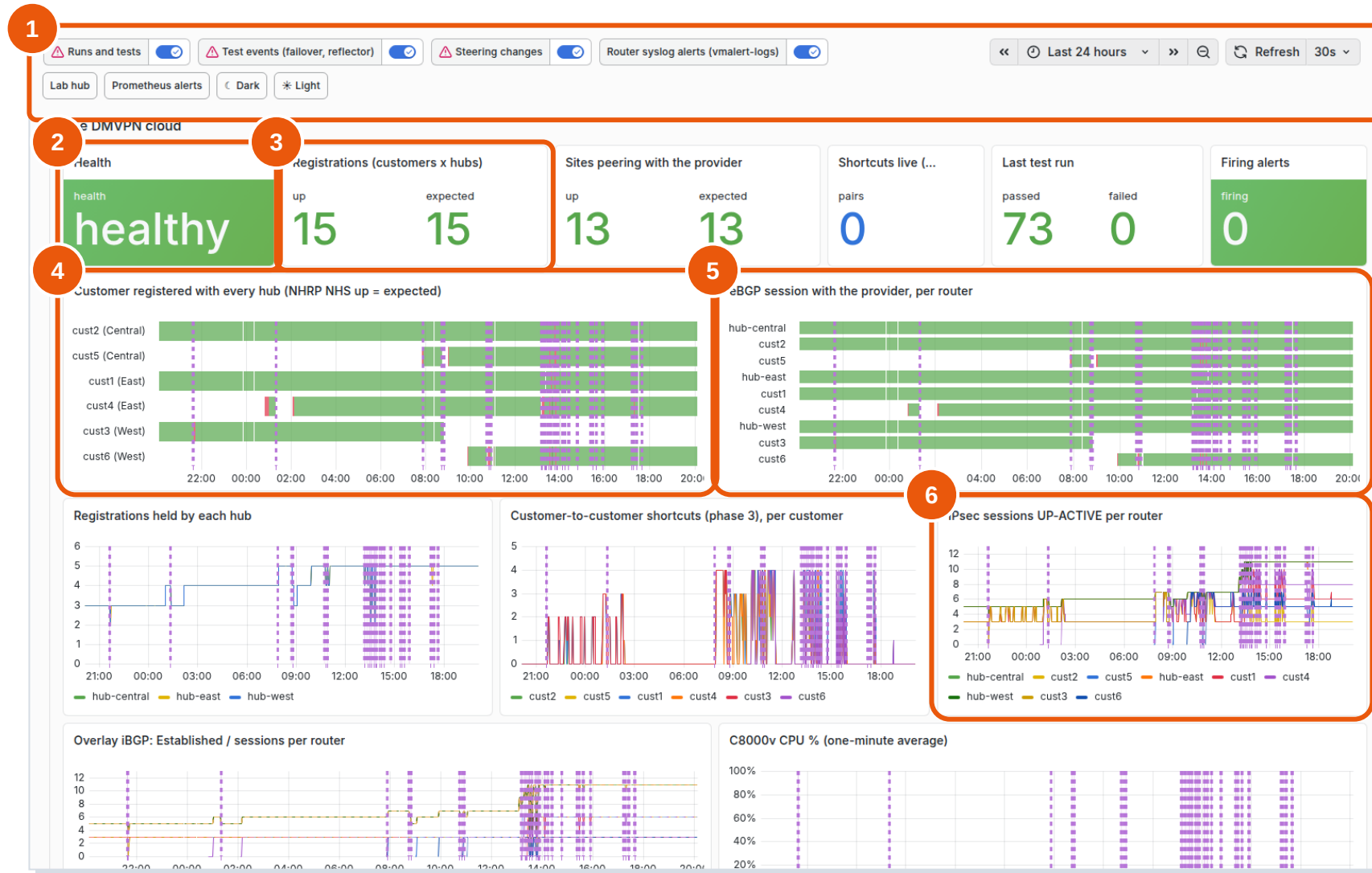
The shared monitoring stack on the NMS covers this lab. IOS XE has no exporter, so the portal itself measures the C8000v routers every minute and publishes the results as Prometheus metrics.



Alerts: DmvpnCustomerNotRegistered, DmvpnProviderSessionDown, DmvpnOverlayBgpDown, DmvpnRouterUnreachable, DmvpnLanHostDown, DmvpnRouterCpuHigh from metrics; BGP neighbour down, NHS down and IKE SA down from syslog.

Maintenance: a job, a simulated failure or declared maintenance mutes the affected nodes' alerts and is left out of the SLA.

Grafana: the DMVPN cloud over the last 24 hours



1 Portal jobs on the timeline
Purple markers: every test run, failover and steering change the portal made.

2 Health
The portal's verdict and firing alerts.

3 Registrations
15 of 15: five customers, three hubs each.

4 Each customer, each hub
Gaps line up with the failover experiments.

5 eBGP to the provider
Per router, over time.

6 IPsec per router
Sessions UP-ACTIVE.

Why it matters

What the portal shows now, Grafana shows over time, next to what the portal was doing.

Grafana: the overlay, the routers and the provider



1 Overlay iBGP
Established vs configured sessions per router.

2 C8000v CPU
Measured by the portal over SSH: IOS XE has no exporter.

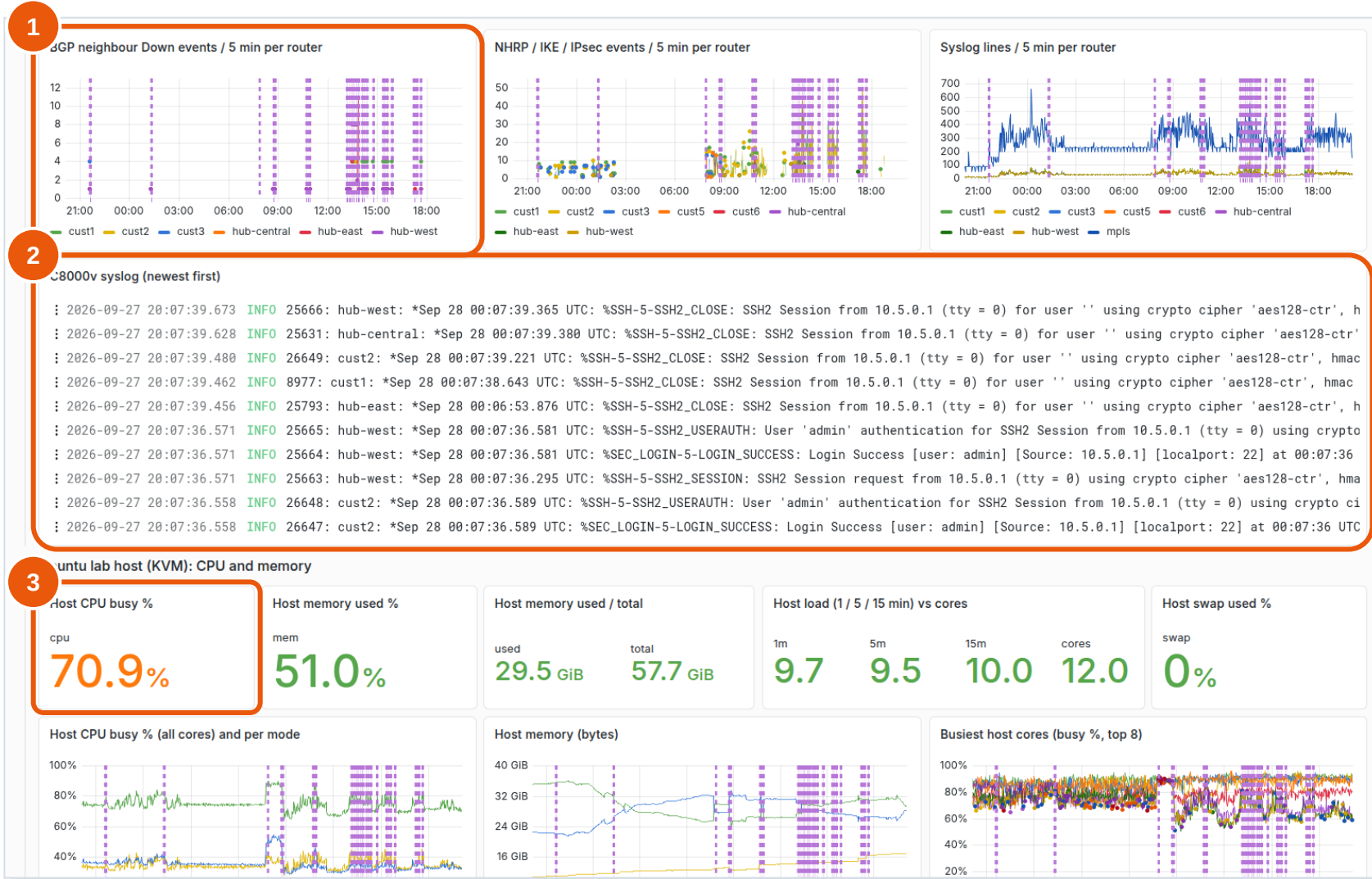
3 The provider
node-exporter, frr-exporter and Telegraf: traffic on every access port.

4 Syslog events
NHRP, IKE and IPsec messages per router, counted in VictoriaLogs.

Why it matters

Underlay and overlay on one page: a provider problem and its DMVPN symptoms line up.

Grafana: router syslog



1 BGP Down from syslog
Neighbour-down messages counted every 5 minutes per router.

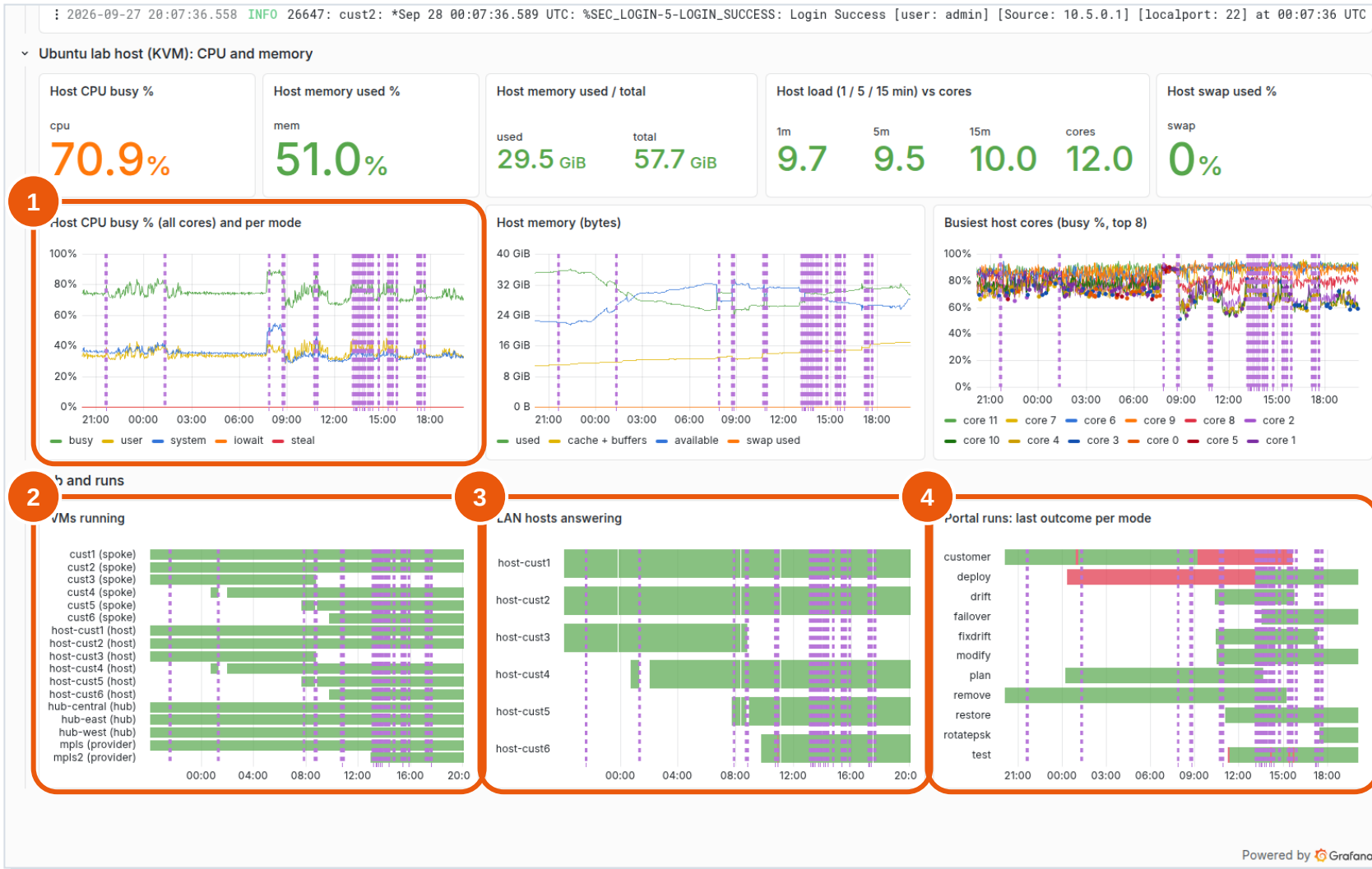
2 Router syslog
Every C8000v line, newest first, from VictoriaLogs.

3 The lab host
CPU, memory, load and swap of the KVM host.

Why it matters

Logs and metrics side by side: no logging in to routers to read their buffers.

Grafana: the lab and the portal's jobs

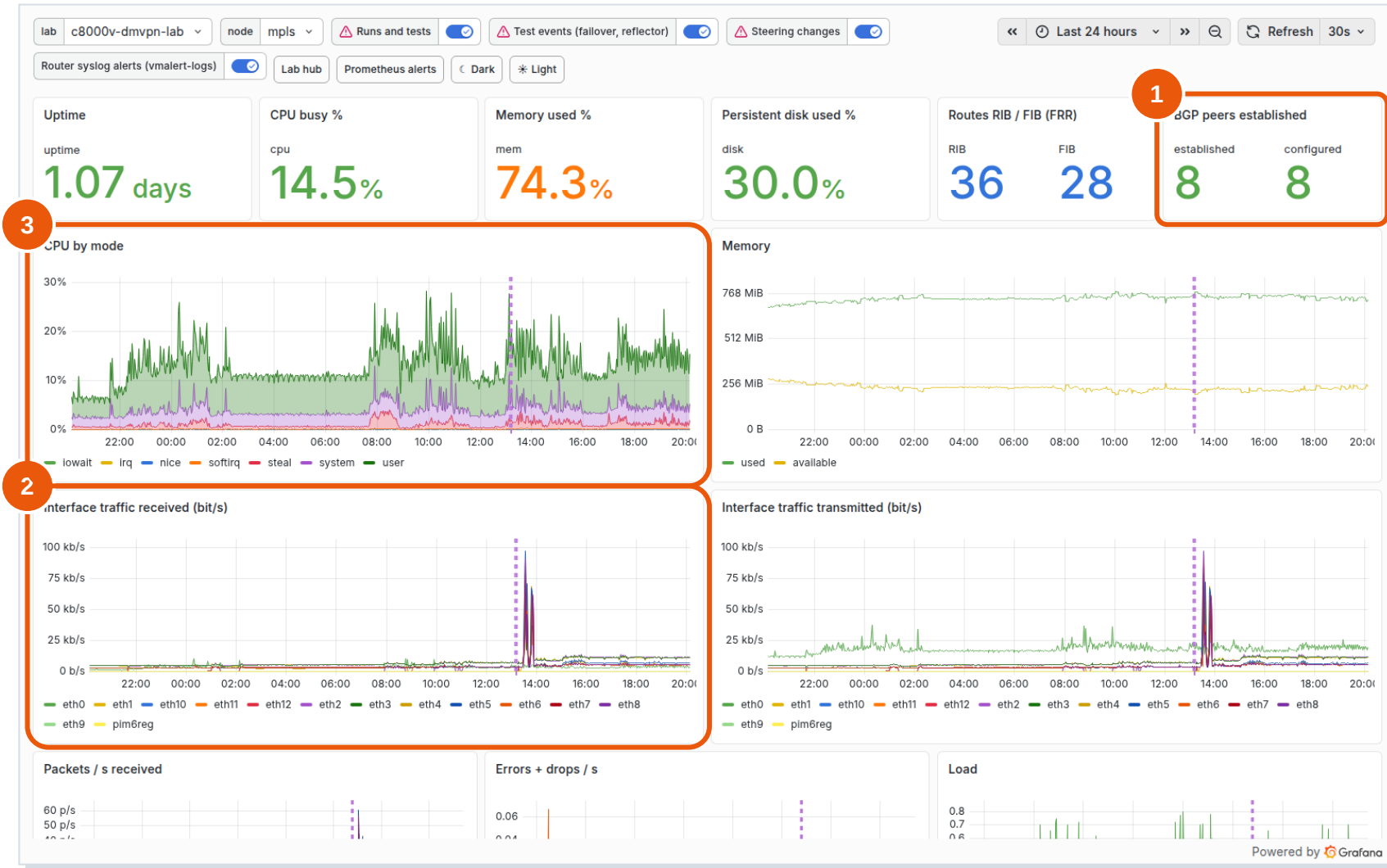


- Host CPU**
All cores, per mode.
- VMs running**
Every router, host and provider VM.
- LAN hosts answering**
Reachability of each site's host.
- The portal's jobs**
The last outcome of each job type; red was a failed run.

Why it matters

Even the automation is monitored: a failing job type shows up like any other fault.

Grafana: the provider router in detail

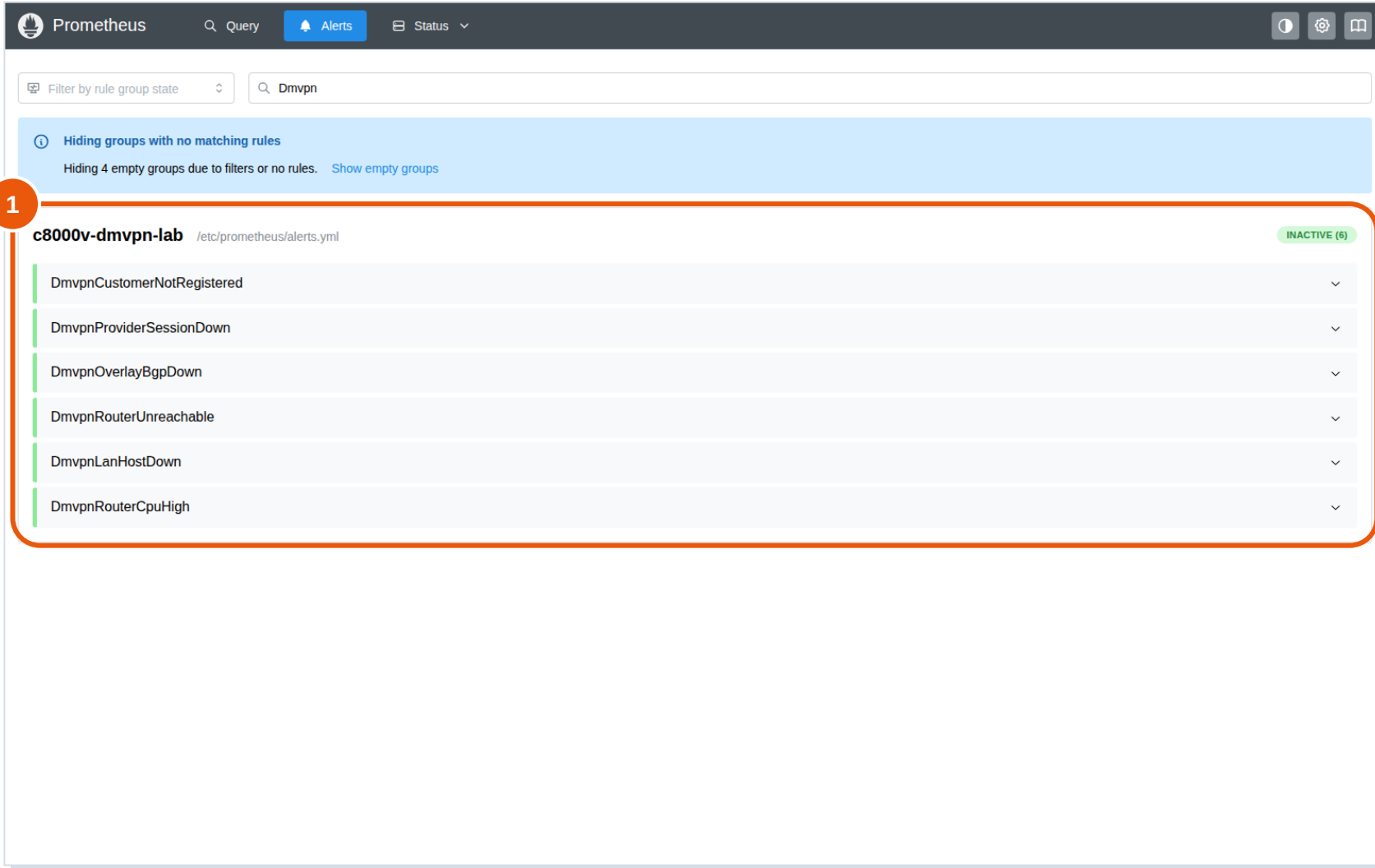


- 1 BGP peers**
8 of 8 established on mpls.
- 2 Access links**
Traffic on every customer and hub port.
- 3 CPU, memory, load**
From node-exporter and Telegraf.

Why it matters

Each VyOS router has full telemetry, down to each interface and BGP peer.

The alert rules



The screenshot shows the Prometheus Alerts interface. At the top, there's a navigation bar with 'Prometheus', 'Query', 'Alerts', and 'Status'. Below this, a search bar contains 'Dmvpn'. A blue banner indicates 'Hiding groups with no matching rules' and 'Hiding 4 empty groups due to filters or no rules.' A red circle with the number '1' highlights the 'c8000v-dmvpn-lab' rule group, which is marked as 'INACTIVE (6)'. This group contains six rules, each with a green status indicator and a dropdown arrow:

- DmvpnCustomerNotRegistered
- DmvpnProviderSessionDown
- DmvpnOverlayBgpDown
- DmvpnRouterUnreachable
- DmvpnLanHostDown
- DmvpnRouterCpuHigh

1

Dmvpn* rules

Six metric rules for this lab, all inactive: healthy. Three more rules watch syslog in vmlert. All are gated on the lab running and muted during maintenance.

Why it matters

Problems page someone before a customer calls, and planned work does not.

Measured, not assumed

3 / 3

hubs every spoke is registered with

20 / 20

LAN hosts reach every other host

33 / 33

application checks passing

73 / 73

Robot tests pass after each change

0.6 s

worst failover when a hub fails; 29 of 35 flows never noticed

6.8 s

worst failover for dual-homed flows when a provider fails

54

IKE SAs re-authenticated with the new pre-shared key

5 m 28 s

key rotation end to end, one router at a time, then verified

Provider failure: dual-homed customers (cust1, cust4) move to cloud 2 over the second provider. Single-homed customers lose that provider by design, and the portal shows it as 'cut off' rather than hiding it.

Hub failure: customers keep two other hubs, live shortcuts need no hub, and routing converges when BGP gives up on the failed hub (hold 9 s).

In short

One source of truth

lab.conf drives the configuration, Nautobot and the portal, and drift is detected against it.

Every change is a job

It is planned, approved, run, verified and recorded with a before-and-after diff per router.

Resilience you can measure

Simulate a failure and get the outage timeline for every flow.

Customers see their own service

SLA, applications, requests and self-service diagnostics, and nothing else.

Secure by default

Roles, four eyes, key rotation on demand, and secrets never shown.